

Principal Component Analysis and Its Applications

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Name	: Nafiul Islam Naim
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Department of Applied Mathematics
University of Dhaka

Abstract

Principal Component Analysis is a widely used dimensionality reduction technique. This study uses 1940–2025 ERA5 reanalysis data, mRMR feature selection, PCA, and six regression models to predict annual Heat Wave Days across Bangladesh's eight administrative divisions. The results reveal a clear west-to-east spatial gradient, with Rajshahi and Khulna experiencing the highest heat-wave occurrence, while no division shows a statistically significant increasing long-term trend. PCA features derived from the Top-20 mRMR-selected variables provide the best overall performance, and Ridge Regression produces the most stable predictions with an average MAE of approximately 3.58 days. SHAP analysis identifies low-level specific humidity at 1000 hPa is the dominant predictor in six divisions, whereas geopotential height, upper-air temperature, and vertical motion are more influential in Rajshahi and Sylhet. The resulting interpretable framework can support regional heat-wave early warning and future climate-risk assessment in Bangladesh.

Contents

Abstract	1
1 Introduction	6
1.1 Context	6
1.2 Objectives	6
2 Principal Component Analysis	8
2.1 Mathematical Foundations of PCA	8
2.1.1 Projection of High-Dimensional Data onto a Direction Vector	8
2.1.2 Variance of the projection	9
2.1.3 Lagrange Multiplier Optimization	10
2.1.4 Eigenvalue Interpretation of Variance Maximization	10
2.2 Mathematical Steps of PCA	11
2.3 Geometric Interpretation of PCA	11
2.3.1 Problem Setup: Real-World Scenario	13
2.3.2 Raw Feature Space Characteristics	13
2.3.3 Visualization of the Rotation	13
2.3.4 Dimensionality Reduction: From 2D to 1D	14
2.3.5 Generalization to High-Dimensional Data	15
2.4 Selection of Principal Components	15
2.4.1 Explained variance ratio	15
2.4.2 Cumulative explained variance	15
2.4.3 Scree plot (elbow method)	15
2.4.4 Kaiser criterion	16
2.5 Advantages and Limitations of PCA	16
2.6 Applications of PCA in Different Fields	18
3 Dimensionality Reduction: Application of Principal Component Analysis	19
3.1 Introduction to Dimensionality Reduction	19
3.2 Feature Extraction Technique	20
3.3 Application of PCA on a Small Dataset	21
4 An Interpretable PCA-Based Machine Learning Framework for Predicting Annual Heat Wave Days in Bangladesh	25
4.1 Introduction	25
4.1.1 Literature Review	25
4.1.2 Objectives	27
4.2 Methodology	27
4.2.1 Study Area	27

4.2.2	Data	29
4.3	Feature Engineering	33
4.3.1	Data Preprocessing	33
4.3.2	Feature Selection	35
4.3.3	Feature Extraction	35
4.4	Machine Learning Model Development	36
4.4.1	Overview of Artificial Neural Network	36
4.4.2	Architecture of MLP	37
4.4.3	Training Hyperparameters	37
4.4.4	Model Evaluation Metrics	39
4.5	Overall Workflow	40
4.6	Results and Discussion	42
4.6.1	Heat Wave Characteristics in Bangladesh	42
4.6.2	Machine Learning Model Performance	45
4.6.3	Model Interpretation Using SHAP	49
5	Conclusion	53
5.1	Summary of the Study	53
5.2	Key Findings	53
5.3	Limitations of the Study	54
5.4	Recommendations for Future Work	54
5.5	Concluding Remarks	54

List of Figures

1.1	Block diagram of PCA	7
2.1	Projection of data points onto a direction vector w	8
2.2	Mathematical steps of PCA	12
2.3	Conceptual illustration of Principal Component Analysis (PCA)	14
2.4	Principal component selection criteria metrics	16
2.5	Applications of PCA across Scientific and Engineering Fields	18
3.1	The trend of classifier performance with the change of dimension	20
3.2	Visualization of Feature Selection and Feature Extraction Processes for Dimensionality Reduction	21
4.1	Map of the study area	28
4.2	ERA5 grid selection framework for Bangladesh. (a) Complete $0.25^\circ \times 0.25^\circ$ ERA5 grid (594 grid points) within the rectangular study domain (88° – 92.5° E, 20.5° – 26.5° N), including both land and surrounding regions. (b) The 197 land-based grid points retained after masking non-land cells, which were used to extract meteorological variables for the eight administrative divisions of Bangladesh	30
4.3	Flowchart for calculating annual HWDs from ERA5 hourly 2 m temperature data	32
4.4	feature engineering workflow	34
4.5	Architecture of MLP	38
4.6	Overall methodological workflow	41
4.7	Spatial distribution of average annual HWDs across bangladesh's divisions . . .	42
4.8	Spatial pattern of heat wave occurrence in Bangladesh (1940–2025)	43
4.9	Annual HWDS by division, 1940–2025, with 5-year moving average (red) . . .	44
4.10	Box plots of annual heat wave day distributions by division (1940–2025). Boxes show interquartile range and median; diamonds show the mean; whiskers extend to $1.5 \times \text{IQR}$; open circles denote outliers.	45
4.11	Comparison of MAE heatmaps for PCA-transformed features selected using mRMR with Top-10, Top-15, and Top-20 features across the eight divisions of Bangladesh.	48
4.12	SHAP summary (beeswarm) plots of the Ridge Regression model for Rajshahi, Khulna, Rangpur, and Dhaka using the Top-20 mRMR-selected features	50
4.13	SHAP summary (beeswarm) plots of the Ridge Regression model for mymensingh, barisal, Chittagong, Sylhet using the Top-20 mRMR-selected features	51

List of Tables

3.1	The summary of dimension reduction technique	20
3.2	Small dataset with three atmospheric features	22
3.3	Mean-centered data matrix $\mathbf{X}_c$	22
3.4	Explained variance ratio per principal component	23
3.5	Final dataset after PCA: two principal components replacing three original features	24
4.1	Description of Bangladesh's eight divisions	28
4.2	Atmospheric variables at different pressure levels	31
4.3	Summary of Study Methodology and Data Parameters	33
4.4	Hyperparameter Settings for the Multi-Layer Perceptron Regressor	39
4.5	Summary statistics of annual HWDs by division, 1940–2025. Trend p-values marked * are statistically significant at the 0.05 level.	44
4.6	Performance of ML models for HWD prediction using PCA (mRMR-10)	46
4.7	Performance of ML models for HWD prediction using PCA (mRMR-15)	47
4.8	Performance of ML models for HWD prediction using PCA (mRMR-20)	47
4.9	Division-wise Summary of Top Contributing Features and SHAP Influence for Heat Wave Day (HWD) Prediction	49

Chapter 1

Introduction

1.1. Context

The rapid growth of data across scientific disciplines has created a need for methods that can reveal meaningful structure while preserving essential information. Modern datasets are becoming larger and more complex. climate stations record numerous atmospheric variables, genomic experiments measure thousands of gene expressions, and digital images consist of huge amounts of pixel data. Extracting useful information from these complex data is difficult without robust dimension reduction techniques. Principal Component Analysis (PCA) is one of the oldest, simplest, and most widely used methods developed to address this problem [1]. The primary objective of PCA is to extract the important information from the data table and to express this information as a set of new orthogonal variables called principal components [2]. By transforming highly correlated features into these orthogonal components, the algorithm highlights the most dominant underlying patterns. Consequently, PCA compresses a high-dimensional dataset into a lower-dimensional representation, making it significantly easier to visualize the data and interpret the sources of variability within the dataset [3]. This foundational methodology was first introduced by Karl Pearson in 1901 as a method for finding the best-fitting lines and planes for a set of points in space [4]. Later, in 1933, Harold Hotelling formalized the method and introduced the term principal components [5]. PCA has become a fundamental tool in multivariate data analysis, widely applied across statistics, machine learning, and scientific research. Its ability to reduce dimensionality while preserving maximum variance makes it an essential technique in modern data-driven studies. The block diagram in Figure 1.1 illustrates how PCA transforms high-dimensional data into a lower-dimensional representation. This combination of theoretical importance and practical usefulness provides the motivation for the objectives of the present study.

1.2. Objectives

Building on this context, the study connects the theoretical understanding of PCA with its application to a real-world climate prediction problem. The main objective of this study is to investigate the theoretical foundations and applications of PCA as an effective dimensionality reduction technique. The study further aims to develop a PCA-based machine learning framework for predicting annual Heat Wave Days (HWDs) in Bangladesh using ERA5 reanalysis atmospheric data. In addition, it seeks to evaluate the effectiveness of PCA in improving prediction performance and to identify, using SHapley Additive exPlanations (SHAP), the dominant atmospheric drivers associated with heat-wave occurrence across Bangladesh. Together,

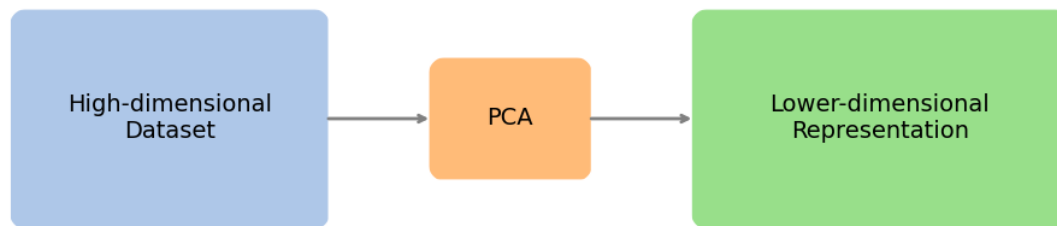


Figure 1.1: Block diagram of PCA

these objectives link the mathematical principles of PCA with an interpretable and practical framework for understanding and predicting heat-wave occurrence in Bangladesh.

Chapter 2

Principal Component Analysis

2.1. Mathematical Foundations of PCA

In this section, we develop the mathematical foundation of PCA). We explain how high-dimensional data are projected onto directions that preserve the maximum possible variance. We then use the covariance matrix, Lagrange multiplier optimization, eigenvalues, and eigenvectors to derive the principal components mathematically.

2.1.1 Projection of High-Dimensional Data onto a Direction Vector

The first step of PCA is to project high-dimensional data onto a single direction [6]. Let $X = [x_1, x_2, \dots, x_n]$ be the data set to be analyzed by PCA, consisting of n observations in a p -dimensional space (e.g. Features) where each observation $x_i \in \mathbb{R}^p$. The goal is to determine the optimal direction vector w onto which the data will be projected [7].

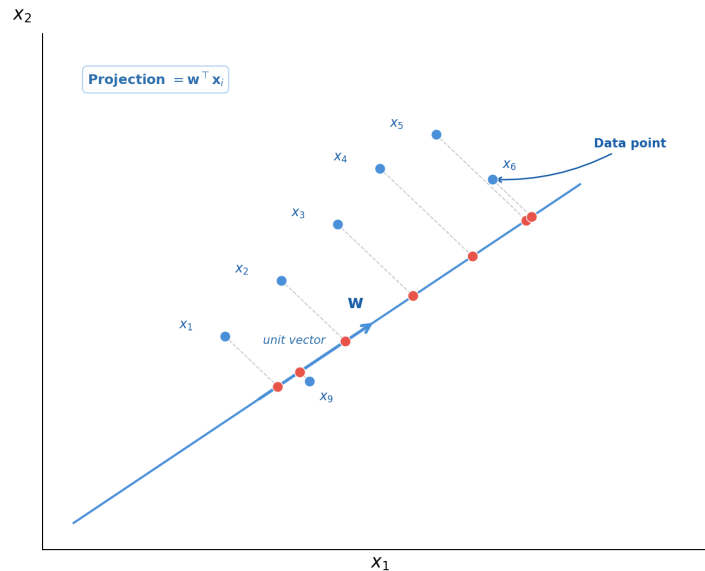


Figure 2.1: Projection of data points onto a direction vector w .

Let w denote a unit direction vector satisfying

$$\|w\| = 1 \tag{2.1}$$

The scalar projection of each observation x_i in the direction w is defined as [1] :

$$z_i = \mathbf{w}^\top \mathbf{x}_i \quad (2.2)$$

where z_i represents the projection of each observation onto the direction vector w . Since z_i is a scalar quantity, each projected observation is represented by a single numerical value. The collection of projected values $z_1, z_2, \dots, z_n$ forms the new one-dimensional representation of the original dataset, as shown in figure 2.1. To ensure that the projection direction does not arbitrarily scale the projected values, the vector w must satisfy the unit norm constraint as follow [1] :

$$\|\mathbf{w}\|^2 = \mathbf{w}^\top \mathbf{w} = 1 \quad (2.3)$$

This constraint is essential because, without normalization, the projection values could be increased indefinitely simply by scaling the vector w . Such scaling would not provide meaningful information about the intrinsic structure of the data. In PCA, the dataset is mean-centered such that $\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = 0$. This ensures the projected data has zero mean and the covariance matrix accurately captures variance [8].

2.1.2 Variance of the projection

Let (Σ) be a real, symmetric $n \times n$ covariance matrix of a data matrix X . The covariance (Σ) and the variance of the projected data can be represented by:

$$\Sigma = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \quad (2.4)$$

$$\text{Var}(z) = \frac{1}{n} \sum_{i=1}^n z_i^2 \quad (2.5)$$

Substituting (2.2) into (2.5) gives

$$\text{Var}(z) = \frac{1}{n} \sum_{i=1}^n (\mathbf{w}^\top \mathbf{x}_i)^2 \quad (2.6)$$

Using the identity

$$(\mathbf{w}^\top \mathbf{x}_i)^2 = (\mathbf{w}^\top \mathbf{x}_i)(\mathbf{w}^\top \mathbf{x}_i)^\top = \mathbf{w}^\top (\mathbf{x}_i \mathbf{x}_i^\top) \mathbf{w} \quad (2.7)$$

Equation (2.6) can be rewritten as

$$\text{Var}(z) = \mathbf{w}^\top \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right) \mathbf{w} \quad (2.8)$$

Therefore, the variance of the projected data becomes

$$\boxed{\text{Var}(z) = \mathbf{w}^\top \Sigma \mathbf{w}} \quad (2.9)$$

The quantity $\mathbf{w}^\top \Sigma \mathbf{w}$ represents the variance of the projected data along the direction w . A larger variance indicates greater data spread and better preservation of information. Therefore, PCA determines the direction w that maximizes the projected variance

$$\boxed{\max \mathbf{w}^\top \Sigma \mathbf{w}} \quad (2.10)$$

subject to the constraint

$$\boxed{\mathbf{w}^\top \mathbf{w} = 1} \quad (2.11)$$

2.1.3 Lagrange Multiplier Optimization

We introduce a Lagrange multiplier λ to handle the constraint $w^T w = 1$

$$\mathcal{L}(w, \lambda) = w^T \Sigma w - \lambda(w^T w - 1) \quad (2.12)$$

Take derivative w.r.t. w

$$\frac{\partial}{\partial w}(w^T \Sigma w) = 2\Sigma w \quad (2.13)$$

$$\frac{\partial}{\partial w}(\lambda w^T w) = 2\lambda w \quad (2.14)$$

So the gradient becomes:

$$\nabla_w \mathcal{L} = 2\Sigma w - 2\lambda w \quad (2.15)$$

Set gradient to zero

$$\Sigma w = \lambda w \quad (2.16)$$

Multiply both sides on the left by w^T

$$w^T \Sigma w = \lambda w^T w = \lambda \cdot 1 = \lambda \quad (2.17)$$

Thus, the variance of the projected data is equal to the eigenvalue.

2.1.4 Eigenvalue Interpretation of Variance Maximization

since (Σ) is real and symmetric, it can be decomposed into its eigenvalues and eigenvectors:

$$\Sigma = \sum_{i=1}^n \lambda_i \mathbf{v}_i \mathbf{v}_i^T \quad (2.18)$$

where the eigenvalues are sorted in descending order

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \quad (2.19)$$

and $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ forms an orthonormal basis, meaning

$$\mathbf{v}_i^T \mathbf{v}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (2.20)$$

Now we can express $\mathbf{w}$ as a linear combination of the orthonormal eigenvectors. since $\mathbf{w}^T \mathbf{w} = 1$

$$\mathbf{w} = \sum_{i=1}^n c_i \mathbf{v}_i \quad (2.21)$$

substitute 2.21 into 2.9 gives

$$\text{Var}(\mathbf{z}) = \mathbf{w}^T \Sigma \mathbf{w} = \mathbf{w}^T \left(\sum_{i=1}^n c_i \mathbf{v}_i \right) \quad (2.22)$$

Using the linear transformation property of eigenvectors, $\Sigma \mathbf{v}_i = \lambda_i \mathbf{v}_i$, we can simplify the expression:

$$\text{Var}(\mathbf{z}) = \mathbf{w}^T \left(\sum_{i=1}^n c_i \lambda_i \mathbf{v}_i \right) \quad (2.23)$$

Substituting the expanded form of $\mathbf{w}^T$ yields:

$$\text{Var}(\mathbf{z}) = \left(\sum_{j=1}^n c_j \mathbf{v}_j^T \right) \left(\sum_{i=1}^n c_i \lambda_i \mathbf{v}_i \right) = \sum_{i=1}^n c_i^2 \lambda_i \quad (2.24)$$

Since λ_1 is defined as the largest eigenvalue ($\lambda_1 \geq \lambda_i$ for all i), we bound the summation from above by replacing each λ_i with λ_1 :

$$\sum_{i=1}^n c_i^2 \lambda_i \leq \sum_{i=1}^n c_i^2 \lambda_1 \quad (2.25)$$

$$\text{Var}(\mathbf{z}) \leq \lambda_1 \sum_{i=1}^n c_i^2 \quad (2.26)$$

the coefficients satisfy the unit vector constraint $\sum_{i=1}^n c_i^2 = 1$

$$\text{Var}(\mathbf{z}) \leq \lambda_1 \quad (2.27)$$

The variance of any projected vector is bounded above by the largest eigenvalue λ_1 . To achieve this maximum bound exactly ($\text{Var}(\mathbf{w}) = \lambda_1$), we must set $c_1 = 1$ and $c_i = 0$ for all $i > 1$. Substituting these coordinate values back into the basis expansion yields:

$$\mathbf{w} = 1\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_n = \mathbf{v}_1 \quad (2.28)$$

Thus, maximizing the projected variance requires selecting the eigenvector $\mathbf{v}_1$ corresponding to the largest eigenvalue λ_1 of the covariance matrix.

2.2. Mathematical Steps of PCA

Figure 2.2 presents the sequential mathematical workflow of PCA. The process begins by centering and standardizing the raw dataset, followed by computing its covariance matrix and performing eigenvalue decomposition. The eigenvalues and eigenvectors are then sorted to select the most informative principal components. Finally, the standardized data are projected onto these components to obtain a reduced-dimensional representation.

2.3. Geometric Interpretation of PCA

To illustrate the geometric interpretation and practical application of PCA, we consider a real-world example based on the morphological analysis of beans. As shown in Figure 2.3, morphological features are extracted from bean images and transformed into a lower-dimensional representation through PCA. The figure illustrates the complete PCA workflow, including feature extraction, the original correlated feature space, coordinate rotation to obtain principal components, and dimensionality reduction. This example demonstrates how PCA transforms the original correlated feature space into orthogonal principal components and projects the data onto a lower-dimensional space while preserving most of the information contained in the original dataset [9].

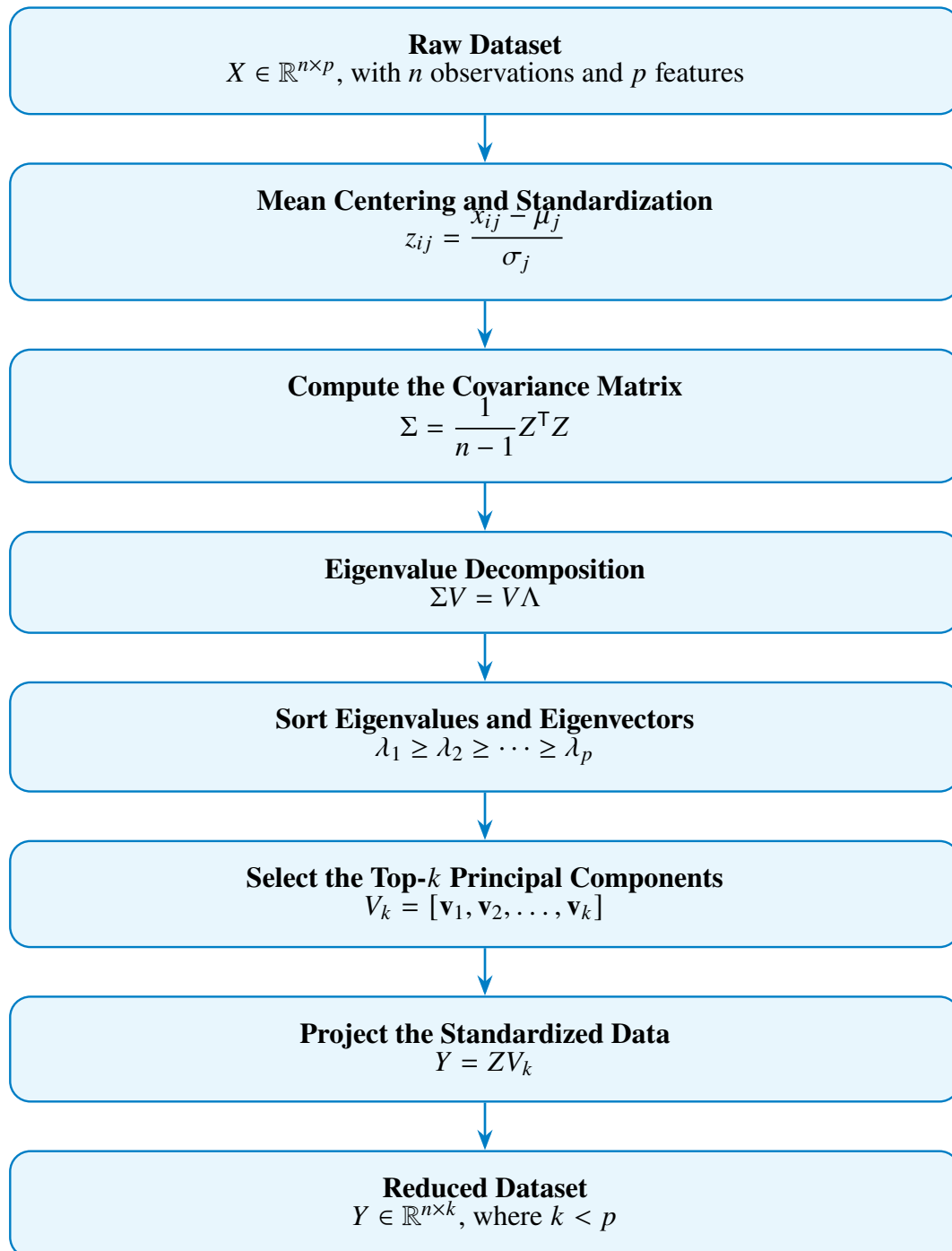


Figure 2.2: Mathematical steps of PCA

2.3.1 Problem Setup: Real-World Scenario

2.3.1.1 Data Collection and Feature Extraction

Consider the task of automatically classifying different types of beans using image analysis (fig:2.3a). Rather than using the raw pixel values from images (which would constitute thousands of dimensions), we extract meaningful morphological features that characterize each object [10]. For each object in the dataset, two key features are computed:

- **Feature 1: Diameter**

The maximum linear dimension of the object, measured in standardized units.

- **Feature 2: Area**

The two-dimensional area enclosed by the object boundary. We use the transformation $\sqrt{\text{Area}}$ rather than raw Area to create a more balanced feature space with comparable scales to diameter measurements.

These features form the basis of our 2D feature space, mathematically represented as:

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} \\ x_{2,1} & x_{2,2} \\ \vdots & \vdots \\ x_{n,1} & x_{n,2} \end{bmatrix}$$

where $x_{i,1}$ represents the diameter and $x_{i,2}$ represents $\sqrt{\text{Area}}$ for the i -th object, and n is the total number of objects in the dataset.

2.3.2 Raw Feature Space Characteristics

The extracted features are plotted in Figure 2.3b. The scatter plot reveals several important characteristics:

- **Strong positive correlation:** Larger objects generally have both larger diameters and larger areas.
- **Non-orthogonal data distribution:** The data are elongated along a diagonal direction rather than being aligned with the coordinate axes.
- **Multicollinearity:** The two features contain overlapping information, motivating the use of Principal Component Analysis (PCA).

2.3.3 Visualization of the Rotation

Figure 2.3c illustrates the effect of PCA on the original feature space shown in Figure 2.3(b). PCA rotates the coordinate system so that the new orthogonal axes align with the directions of maximum and minimum variance. The first principal component (PC1) captures the largest variation in the data, while the second principal component (PC2) captures the remaining variance subject to being orthogonal to PC1. This transformation decorrelates the original variables and converts the covariance matrix into a diagonal matrix whose diagonal elements are the variances (eigenvalues) of the principal components.

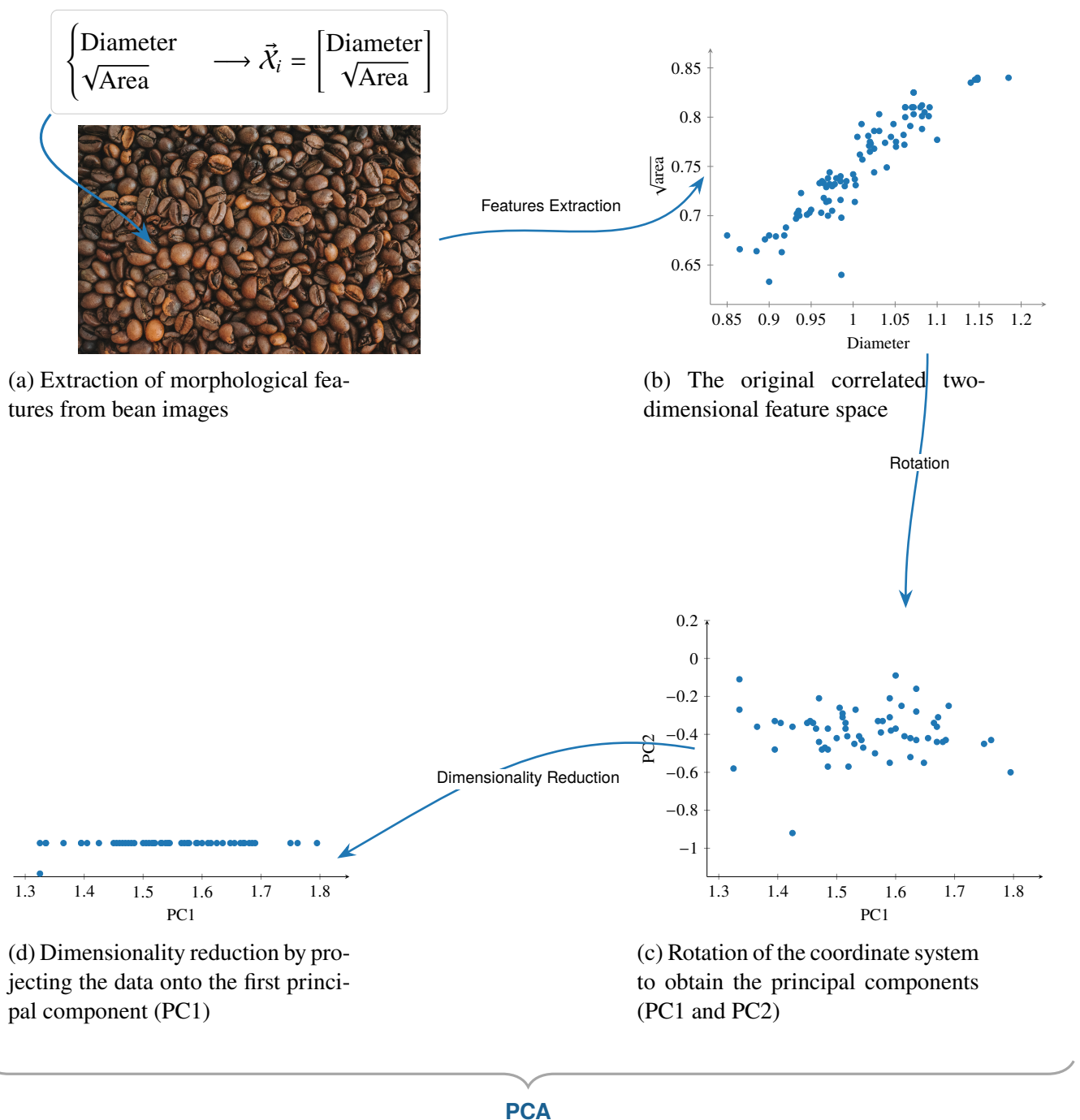


Figure 2.3: Conceptual illustration of Principal Component Analysis (PCA)

2.3.4 Dimensionality Reduction: From 2D to 1D

After the rotation, the data are projected onto the first principal component, as shown in Figure 2.3d. This projection reduces the dimensionality from two variables to a single variable while preserving most of the information contained in the original dataset. Although a small amount of variance associated with PC2 is discarded, the dominant data structure remains largely unchanged. Consequently, dimensionality reduction simplifies computation, reduces storage

requirements, improves numerical stability, and can help reduce overfitting.

2.3.5 Generalization to High-Dimensional Data

While the beans example uses 2D data, the geometric principles extend to arbitrarily high-dimensional datasets[11] : For p -dimensional data, PCA yields p principal components with corresponding eigenvalues, preserving the same orthogonal transformation framework. In practice, most real high-dimensional datasets show rapid variance decay, meaning only a small number of components capture the majority (often >95%) of the total variance. This allows effective dimensionality reduction while retaining most information. As a result, PCA helps mitigate the curse of dimensionality by revealing a lower-dimensional manifold that the high-dimensional data approximately lies on [12].

2.4. Selection of Principal Components

A principal component is a new variable formed as a linear combination of the original variables [1]:

$$PC_1 = w_{11}X_1 + w_{12}X_2 + \cdots + w_{1p}X_p \quad (2.29)$$

where, w_{ij} represents the loadings (weights) and X_i denotes the standardized features. PC1 captures the maximum possible variance. PC2 captures the maximum remaining variance while being orthogonal to PC1 [5, 13]. Subsequent components follow the same rule. To determine the optimal number of principal components, various criteria can be employed, such as explained variance ratio(fig:2.4a), Cumulative explained variance(fig:2.4b), Scree plot (fig:2.4c), and Kaiser criterion(fig:2.4d).

2.4.1 Explained variance ratio

Shows how much of the total variance in your data is captured by each principal component individually. PC1 explains the most; subsequent PCs explain progressively less. Mathematically:

$$\text{Explained Variance Ratio } (EV_i) = \frac{\lambda_i}{\sum_{j=1}^p \lambda_j} \quad (2.30)$$

where, λ_i is the eigenvalue of the i -th principal component, and $\sum_{j=1}^p \lambda_j$ represents the sum of all eigenvalues (total variance).

2.4.2 Cumulative explained variance

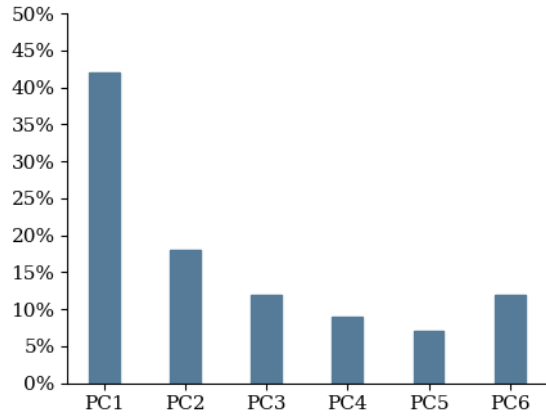
Cumulative variance measures the total variance explained by the first k components.

$$\text{Cumulative Variance}_k = \sum_{i=1}^k \text{Explained Variance Ratio}_i \quad (2.31)$$

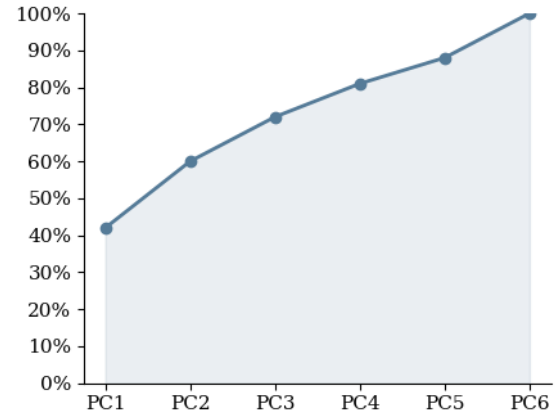
where k represents the number of retained principal components.

2.4.3 Scree plot (elbow method)

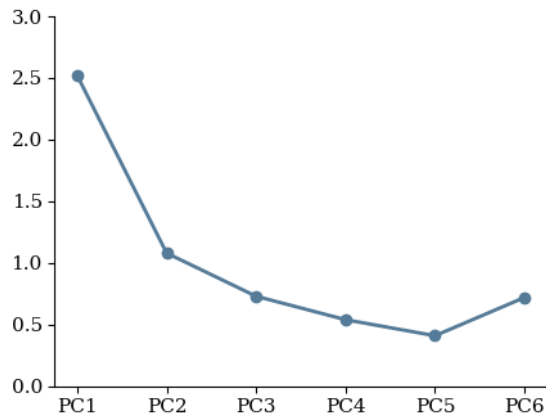
A scree plot displays the eigenvalues against the component number. The plot helps identify the elbow point, where the rate of variance gain drops sharply. The scree plot directly visualizes diminishing returns. Once the curve levels off, additional PCs add little new information.



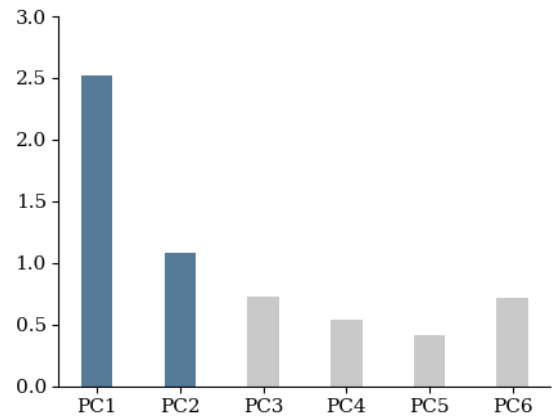
(a) Distribution of explained variance across principal components.



(b) Cumulative explained variance across principal components



(c) Scree Plot



(d) Component retention based on the Kaiser criterion ($\lambda_i > 1$).

Figure 2.4: Principal component selection criteria metrics

2.4.4 Kaiser criterion

The Kaiser criterion states: Retain components with eigenvalues greater than 1.

$$\lambda_i > 1 \quad (2.32)$$

where λ_i represents the eigenvalue of the i -th principal component.

2.5. Advantages and Limitations of PCA

Advantages of PCA

a) Dimensionality Reduction

PCA reduces the complexity of datasets by projecting data onto the first few principal components, while the first principal component captures the maximum variance and subsequent components capture maximum remaining variance [13]. PCA is often used before applying

machine learning algorithms to reduce dimensionality, speed up training and improve model generalization.

b) Eliminates Multicollinearity

In many datasets, features are highly correlated (multicollinear), which can cause issues for algorithms like Linear Regression. PCA transforms these correlated variables into an entirely orthogonal and uncorrelated coordinate system.

c) Data Visualization Capabilities

PCA reduces high-dimensional data to a two-dimensional or three-dimensional space for easier visualization, helping to reveal data trends, clusters or outliers. The PCA algorithm can reduce the dimensionality of data to provide a clearer understanding and visualization of multidimensional data display.

d) Computational Efficiency and Speed

The PCA algorithm is fast and does not require parameter tuning or optimization. PCA is computationally efficient and provides an interpretable transformation in lower dimensions.

e) Reduces noise and redundancy

It can filter out noise by discarding components that capture only low variance.

Limitations of PCA

a) Computational Challenges in Large-Scale Datasets

Although PCA is an effective dimensionality reduction technique, it faces computational challenges when applied to large and high-dimensional datasets [14]. The method relies on eigenvalue decomposition (EVD) [15] or singular value decomposition (SVD) [16], which can be computationally expensive and require substantial memory resources. These limitations become more significant in applications involving large-scale or continuously generated data.

b) Affected by Outliers

PCA is based on the empirical covariance matrix of the data and hence is highly sensitive to outlying observations [17]. Even a single extreme point can significantly change the PCA result and hide the true data structure.

c) Unsupervised Nature

PCA is an unsupervised technique and does not consider the target variable during feature extraction.

d) Fails on Complex, Non-Linear Data

PCA is a linear technique for dimensionality reduction and may fail to represent nonlinear structures [18]. If your data has complex, non-linear relationships, PCA will perform poorly. In those cases, non-linear methods like t-SNE [19] or Kernel PCA [20] are required.

e) Components are difficult to interpret

After transformation, the new principal components are combinations of original features. This fundamentally destroys the physical meaning of the original features, making it highly difficult to explain feature significance to non-technical stakeholders.

2.6. Applications of PCA in Different Fields

PCA is widely used across many scientific and engineering fields because it reduces dimensionality, removes redundancy, and helps in extracting meaningful patterns from high-dimensional data. In several applications of PCA, such as image classification [21], image compression [22], face recognition [23], industrial process modelling [24], quantitative finance [25], neuroscience [26], weather and climate science [27], machine learning and data science [28], genetics and genomics [29], it plays an important role in simplifying complex datasets while preserving essential information. This wide applicability makes PCA one of the most fundamental techniques in modern data analysis and machine learning. Figure 2.5 represents the Scientific Applications of PCA.

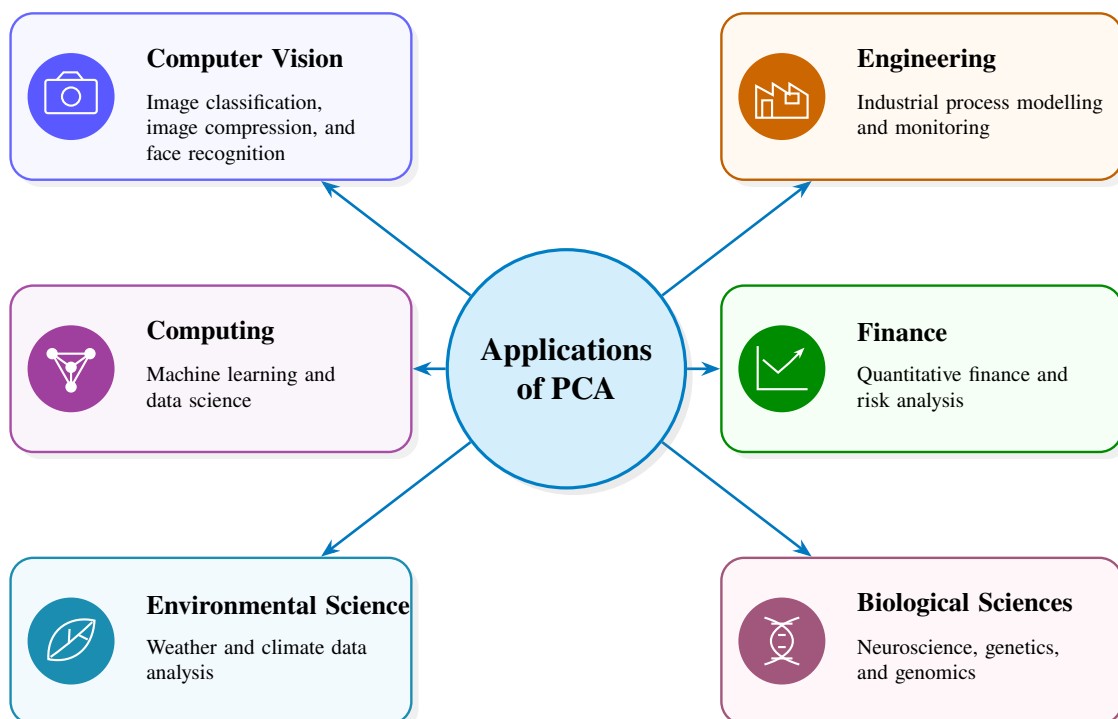


Figure 2.5: Applications of PCA across Scientific and Engineering Fields

Chapter 3

Dimensionality Reduction: Application of Principal Component Analysis

3.1. Introduction to Dimensionality Reduction

In modern data science, machine learning, and computational biology, datasets often contain hundreds or even thousands of features (variables). While high-dimensional data provide rich and detailed information, they also introduce significant challenges for data analysis and predictive modelling [30]. As the number of features increases, data become increasingly sparse in the feature space, distances between observations become less informative, computational costs rise substantially, and machine learning models become more prone to overfitting. This phenomenon is commonly referred to as the "curse of dimensionality", a term introduced by Richard Bellman in the context of dynamic programming [31]. As shown in Figure 3.1, classifier accuracy peaks at an optimal number of dimensions before steadily degrading as more features are introduced. Dimensionality Reduction (DR) is the process of transforming high-dimensional data into a lower-dimensional representation while preserving as much relevant information, geometric structure, and variance as possible [18]. By reducing the number of variables, dimensionality reduction techniques can improve computational efficiency, enhance model generalization, facilitate data visualization, and reduce noise. Most dimensionality reduction methods are motivated by the "manifold hypothesis", which states that high-dimensional observations often lie on or near a lower-dimensional manifold embedded within the original feature space. Consequently, the intrinsic dimensionality of the data is typically much smaller than its observed dimensionality, making it possible to represent complex datasets using a smaller set of informative features while retaining their essential structure [32]. Dimensionality reduction techniques are generally divided into two main categories: feature selection [33] and feature extraction [34]. In feature selection, some information may be lost because certain features are excluded during the process of choosing a subset of the original variables, which can reduce the overall information available for modelling. In contrast, feature extraction reduces dimensionality by transforming the original features into a new set of lower-dimensional representations, often preserving most of the important information from the initial dataset while achieving a more compact form [18, 35]. Table 3.1 provides a descriptive summary of the methods of dimension reduction. Suppose a dataset contains the features : {Height, Weight, Age, Income, Education}. Feature Selection : {Height, Age, Education}, Feature Extraction : {PC1, PC2, PC3}, where each principal component is a combination of all original features. Therefore, Feature selection reduces dimensionality by retaining only the most relevant original features, whereas feature extraction transforms the original feature space into a new lower-

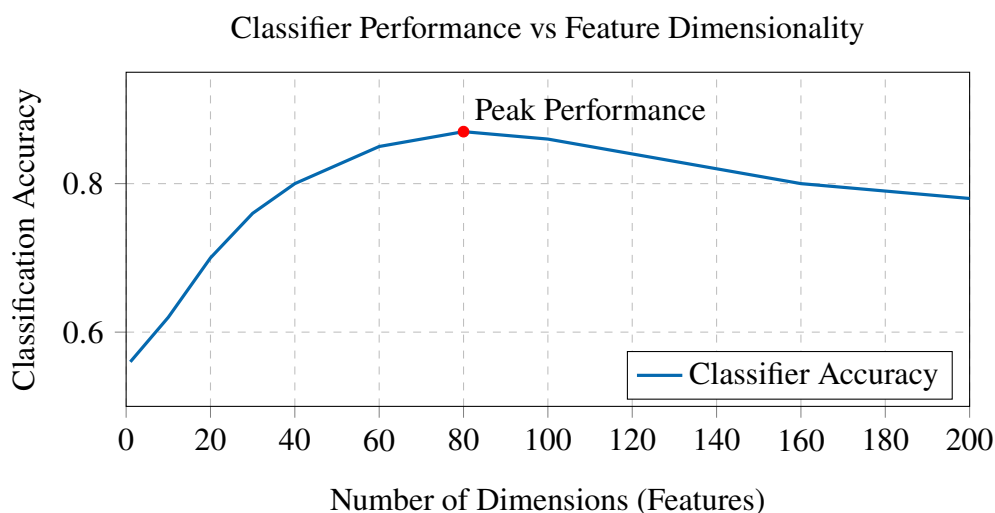


Figure 3.1: The trend of classifier performance with the change of dimension

Table 3.1: The summary of dimension reduction technique

Method	Feature Selection	Feature Extraction
Key concept	Selects a subset of relevant features from the original set of features	Summarize the dataset by creating linear combinations of the original features
Data Transformation	Retains the original features.	Original features are transformed into a new feature space.
Information Loss	May lose important information if relevant features are removed.	Attempts to preserve most of the information while reducing dimensions.
Goal	Remove redundant, irrelevant, or noisy features	Compress information into a smaller set of informative features.

dimensional representation. Feature selection preserves interpretability, while feature extraction aims to maximize information retention and data compression. Fig:3.2 depicts the Visualization of Feature Selection and Feature Extraction Processes.

3.2. Feature Extraction Technique

The commonly used feature extraction methods include PCA [1], Independent Component Analysis (ICA) [36], Linear Discriminant Analysis (LDA) [37], and neural network-based methods, such as Multilayer Perceptrons (MLPs) [38], are among the most widely used. Among these, PCA and ICA are statistical techniques that extract features while preserving either the variance in the data (PCA) or the mutual information among variables (ICA). These methods are generally categorized as unsupervised approaches, as they do not utilize class label information during feature transformation. Consequently, the extracted features may not always be optimal for classification tasks. Recent advances in machine learning have further expanded the scope of feature extraction through deep learning architectures, including autoencoders and deep neural networks. These methods automatically learn hierarchical feature representations from

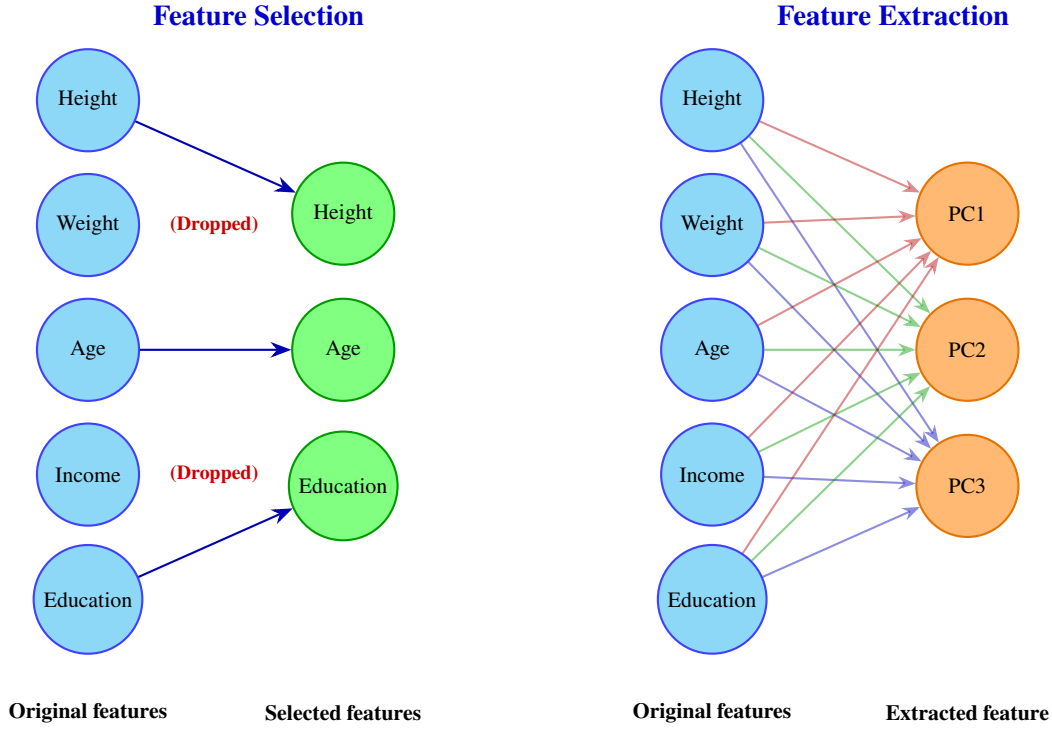


Figure 3.2: Visualization of Feature Selection and Feature Extraction Processes for Dimensionality Reduction

raw data and have demonstrated remarkable success in image recognition, natural language processing, and pattern classification tasks [39, 40]. Nevertheless, classical methods such as PCA, ICA, and LDA remain widely adopted due to their simplicity, interpretability, and computational efficiency [41]. Overall, feature extraction techniques play a crucial role in reducing dimensionality and improving learning efficiency. The choice of an appropriate method depends on the characteristics of the dataset, the availability of class labels, computational constraints, and the objectives of the learning task.

3.3. Application of PCA on a Small Dataset

To illustrate the mechanics of PCA before applying it to the full ERA5-derived feature set, we work through a small, self-contained numerical example. Consider a synthetic weather dataset consisting of $n = 6$ daily observations described by $p = 3$ features: Temperature (T , $^{\circ}\text{C}$), Relative Humidity (H , $\%$), and Wind Speed (W , m s^{-1}). The objective is to reduce these three correlated features to two uncorrelated principal components, PC_1 and PC_2 , while retaining as much of the original variance as possible.

Step 1: The Raw Dataset

The raw data matrix $\mathbf{X} \in \mathbb{R}^{6 \times 3}$ is given in Table 3.2.

Step 2: Mean Centering

PCA requires the data to be mean-centered so that each feature has zero mean. The mean vector $\bar{\mathbf{x}}$ is computed column-wise as

Table 3.2: Small dataset with three atmospheric features

Day	T ($^{\circ}\text{C}$)	H (%)	W (m s^{-1})
1	30	55	10
2	32	50	6
3	28	65	9
4	35	45	4
5	31	60	12
6	29	58	7

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = [30.833 \quad 55.500 \quad 8.000]. \quad (3.1)$$

Subtracting $\bar{\mathbf{x}}$ from every row of $\mathbf{X}$ gives the centered matrix $\mathbf{X}_c = \mathbf{X} - \bar{\mathbf{x}}$ (Table 3.3).

Table 3.3: Mean-centered data matrix $\mathbf{X}_c$

Day	$T - \bar{T}$	$H - \bar{H}$	$W - \bar{W}$
1	-0.833	-0.500	2.000
2	1.167	-5.500	-2.000
3	-2.833	9.500	1.000
4	4.167	-10.500	-4.000
5	0.167	4.500	4.000
6	-1.833	2.500	-1.000

Step 3: Covariance Matrix

The 3×3 sample covariance matrix is computed as

$$\mathbf{C} = \frac{1}{n-1} \mathbf{X}_c^{\top} \mathbf{X}_c, \quad (3.2)$$

which, for $n = 6$, evaluates to

$$\mathbf{C} = \begin{bmatrix} 6.167 & -16.100 & -4.200 \\ -16.100 & 51.500 & 15.400 \\ -4.200 & 15.400 & 8.400 \end{bmatrix}. \quad (3.3)$$

The large off-diagonal magnitudes (e.g. $\text{Cov}(T, H) = -16.100$) indicate strong linear correlation between the original features — this redundancy is precisely what PCA exploits to reduce dimensionality.

Step 4: Eigenvalues and Eigenvectors

The principal directions of $\mathbf{C}$ are found by solving the characteristic equation

$$\det(\mathbf{C} - \lambda \mathbf{I}) = 0, \quad (3.4)$$

which yields a cubic polynomial in λ . Solving this equation (via eigen-decomposition) gives three eigenvalues, sorted in descending order, and their corresponding eigenvectors:

$$\lambda_1 = 61.433, \quad \lambda_2 = 3.759, \quad \lambda_3 = 0.875, \quad (3.5)$$

$$\mathbf{v}_1 = \begin{bmatrix} -0.288 \\ 0.913 \\ 0.288 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0.306 \\ -0.197 \\ 0.931 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0.907 \\ 0.356 \\ -0.223 \end{bmatrix}. \quad (3.6)$$

Each eigenvector $\mathbf{v}_k$ is normalized so that $\|\mathbf{v}_k\| = 1$, and λ_k represents the amount of variance captured along the direction $\mathbf{v}_k$. As a check, $\lambda_1 + \lambda_2 + \lambda_3 = 66.067$, which equals the trace of $\mathbf{C}$ (i.e. the total variance of the original data), confirming that no variance is lost in the decomposition.

Step 5: Explained Variance and Component Selection

The proportion of total variance explained by each principal component is

$$\text{EVR}_k = \frac{\lambda_k}{\sum_{j=1}^p \lambda_j} \times 100\%, \quad (3.7)$$

giving the results in Table 3.4. Since PC_1 and PC_2 together retain 98.675% of the total variance

Table 3.4: Explained variance ratio per principal component

Component	Eigenvalue λ_k	Explained Variance (%)	Cumulative (%)
PC_1	61.433	92.986	92.986
PC_2	3.759	5.690	98.675
PC_3	0.875	1.325	100.000

in the original three features, we discard PC_3 (which contributes only 1.325%) and reduce the dataset from $\mathbb{R}^3$ to $\mathbb{R}^2$. The projection matrix is formed from the first two eigenvectors:

$$\mathbf{W} = [\mathbf{v}_1 \quad \mathbf{v}_2] = \begin{bmatrix} -0.288 & 0.306 \\ 0.913 & -0.197 \\ 0.288 & 0.931 \end{bmatrix} \in \mathbb{R}^{3 \times 2}. \quad (3.8)$$

Step 6: Projection onto the Principal Components

The reduced-dimension representation $\mathbf{Y} \in \mathbb{R}^{6 \times 2}$ is obtained by projecting the centered data onto $\mathbf{W}$:

$$\mathbf{Y} = \mathbf{X}_c \mathbf{W}. \quad (3.9)$$

Carrying out this matrix multiplication for each of the six observations gives the final PC_1 , PC_2 scores in Table 3.5.

Table 3.5: Final dataset after PCA: two principal components replacing three original features

Day	PC_1	PC_2
1	0.359	1.706
2	-5.935	-0.421
3	9.780	-1.809
4	-11.942	-0.379
5	5.214	2.889
6	2.523	-1.986

Interpretation

The original three-feature dataset $\mathbf{X} \in \mathbb{R}^{6 \times 3}$ has been transformed into a two-feature dataset $\mathbf{Y} \in \mathbb{R}^{6 \times 2}$ while preserving 98.675% of the total variance. Examining the loadings in $\mathbf{v}_1$, PC_1 is dominated by a strong positive contribution from Humidity (0.913) and a comparable positive contribution from Wind Speed (0.288) opposed by Temperature (-0.288), suggesting that PC_1 primarily represents a combined moisture-and-wind axis of variability. PC_2 is dominated by Wind Speed (0.931), capturing residual variability not explained by the temperature-humidity relationship. This simple example reflects the same two-stage approach applied to the ERA5 atmospheric dataset in Chapter 4, where mRMR feature selection is followed by PCA to compress dozens of correlated pressure-level variables into a small set of principal components (e.g. `temp_PC1`, `shum_PC1`, `uwnd_PC1`) prior to regression modeling.

Chapter 4

An Interpretable PCA-Based Machine Learning Framework for Predicting Annual Heat Wave Days in Bangladesh

4.1. Introduction

Bangladesh is among the countries most vulnerable to rising temperatures, and in recent decades the frequency and intensity of extreme heat events across its divisions have drawn growing scientific and public attention. Reliable prediction of annual Heat Wave Days (HWDs) is essential for early warning systems, agricultural planning, and public health preparedness, yet the atmospheric processes that govern heat wave occurrence are complex and involve a large number of interrelated variables such as temperature, humidity, geopotential height, and wind fields at multiple pressure levels. Building a predictive model directly from this high-dimensional atmospheric data is challenging, since many of these variables are strongly correlated with one another, and using all of them simultaneously can introduce redundancy, increase computational cost, and reduce the interpretability of the resulting model. This chapter extends the theoretical treatment of PCA presented in Chapters 2 and 3 to a practical, real-world application: the prediction of annual HWDs across Bangladesh’s eight administrative divisions using ERA5 reanalysis atmospheric data. The chapter combines minimum Redundancy Maximum Relevance (mRMR) feature selection with PCA-based dimensionality reduction to compress a large set of candidate atmospheric predictors into a smaller number of informative, decorrelated components. Six regression models are then trained and compared to identify which approach best predicts HWDs at the divisional level, and SHAP is subsequently used to interpret the physical drivers underlying the model predictions. In this way, the chapter demonstrates how PCA, introduced purely as a mathematical and statistical technique in earlier chapters, can be embedded within a complete, interpretable machine learning pipeline to address a pressing real-world climate problem.

4.1.1 Literature Review

The theoretical foundations of PCA trace back to Karl Pearson, who first introduced the idea of fitting lines and planes of closest fit to a system of points as a means of summarizing multivariate data with minimal loss of information [4]. This geometric formulation was later connected to matrix algebra by Eckart and Young, who proved that the best low-rank approximation of a matrix, in the least-squares sense, is obtained by truncating its singular value decomposition,

a result that underlies the modern computational implementation of PCA [42]. Tipping and Bishop subsequently reformulated PCA within a probabilistic framework, showing that principal components can be derived as the maximum-likelihood solution of a latent variable model with Gaussian noise, which allowed PCA to be extended naturally to missing-data scenarios and mixture models [43]. More recently, Gewers et al. provided a comprehensive and accessible synthesis of PCA theory, standardization practices, visualization strategies, and its application across diverse scientific domains, reflecting the continued relevance of the technique nearly a century after its introduction [44]. The versatility of PCA as a general-purpose dimensionality reduction and pattern-recognition tool is well illustrated by Turk and Pentland's Eigenfaces method, in which PCA was applied to a database of facial images to construct a low-dimensional "face space" capable of efficient face detection and recognition, a landmark application that demonstrated PCA's power well beyond its original statistical setting [11]. Within the atmospheric and climate sciences specifically, PCA has been used both to explore relationships among weather variables and to reduce the dimensionality of meteorological datasets prior to modelling. Manandhar et al. applied PCA to a set of weather and time-related variables, including relative humidity, solar radiation, temperature, and dew point, to study their interdependency and to distinguish rain from no-rain conditions, finding that a small number of principal components could capture the majority of the variance in the original variable set [45]. In a similar spirit, Tadić et al. used PCA to assess the relative importance of ten meteorological parameters and climate change indicators at two climatically contrasting stations, demonstrating how PCA can distill a large indicator set into a smaller number of components that summarize the dominant modes of climate variability [46]. These studies collectively motivate the use of PCA in the present work as a means of condensing a large set of correlated atmospheric predictors, drawn from multiple pressure levels, into a compact set of components suitable for regression modelling. A separate and growing body of literature has focused specifically on the prediction and physical explanation of heat waves using machine learning. Asadollah et al. applied several advanced machine learning models to predict heat waves using meteorological variables across diverse regions of Iran, demonstrating that data-driven models can achieve strong predictive performance for heat wave characteristics when supplied with appropriate atmospheric predictors [27]. Buschow et al. took an explicitly explainable approach, combining a logistic variant of PCA to compress binary heatwave fields with a neural network and Shapley-value-based feature attribution to quantify how much information on European heatwave occurrence could be learned from individual ERA5 variables, an approach conceptually related to the mRMR-PCA-SHAP pipeline adopted in this study [47]. Closer to the present study area, Chaki et al. used the Weather Research and Forecasting (WRF) model to simulate a specific heat wave event over Bangladesh and to analyze the resulting heat stress index, providing valuable physical insight into the atmospheric conditions associated with Bangladeshi heat waves, albeit through a process-based rather than a statistical or machine-learning approach [48]. Bochenek and Ustrnul reviewed several hundred recent studies applying machine learning to weather prediction and climate analysis, concluding that machine learning methods, particularly random forests, support vector machines, and deep learning architectures, are increasingly central to numerical weather prediction and climate research, which supports the model comparison strategy adopted in this chapter [49]. Neumann et al., working in the context of energy time-series forecasting, systematically evaluated different transformations of weather input data, including dimensionality reduction techniques, and found that appropriately transformed weather inputs, rather than raw station or grid data, consistently improved forecast accuracy, reinforcing the rationale for applying feature selection and PCA prior to model training in this study [50]. Finally, Kan et al. developed explainable machine learning classifiers for seasonal heatwave

forecasting using remote sensing and atmospheric features, employing SHAP to reveal how the relative importance of atmospheric versus landscape features shifted with forecast lead time, an interpretability strategy closely mirrored in the SHAP-based analysis presented later in this chapter [51]. Taken together, this literature establishes both the mathematical legitimacy of PCA as a dimensionality reduction technique and its demonstrated usefulness in meteorological and heat wave applications, while also highlighting a growing emphasis on model interpretability through techniques such as SHAP. However, relatively few studies have combined mRMR-based feature selection, PCA-based dimensionality reduction, multiple regression model benchmarking, and SHAP-based physical interpretation within a single, division-level framework for a climatically diverse country such as Bangladesh. This chapter addresses that gap by developing and evaluating such a framework for predicting annual Heat Wave Days across Bangladesh's eight administrative divisions.

4.1.2 Objectives

The specific objectives of this chapter are as follows.

1. To investigate the spatial and temporal characteristics of annual HWDs across Bangladesh.
2. To evaluate the effectiveness of PCA for improving machine learning-based prediction of annual HWDs.
3. To identify the dominant atmospheric factors associated with heat-wave occurrence across the administrative divisions of Bangladesh.

4.2. Methodology

4.2.1 Study Area

The present study focuses on Bangladesh, a South Asian country situated between 20°34'–26°38' N latitude and 88°01'–92°41' E longitude. The country experiences three major seasons: the hot pre-monsoon season (March–May), the monsoon season (June–October), and the dry winter season (November–February). Due to its geographical location and climatic conditions, Bangladesh is highly vulnerable to extreme weather events, including heat waves. Bangladesh is recognized as a climate change hotspot [52] due to its geographical location at the transition between temperate and semi-humid climatic zones. This unique setting makes the country highly vulnerable to changes in large-scale atmospheric circulation patterns. Consequently, Bangladesh is increasingly exposed to climate-related hazards, including extreme heat events. Despite the growing concern over heat stress, only a limited number of studies have examined its impacts within the country, with most existing research focusing on local or regional scales [53]. The increasing frequency and severity of heatwaves have made Bangladesh a suitable case study for investigating regional heatwave characteristics and developing predictive models. To facilitate a comprehensive spatial assessment, the country was divided into its eight administrative divisions: Dhaka, Chittagong, Rajshahi, Khulna, Barishal, Sylhet, Rangpur, and Mymensingh, depicted in Fig:4.1. Table 4.1 presents the description of each area. These divisions represent distinct climatic and geographical settings, allowing for the evaluation of regional variability in heatwave occurrence and intensity.

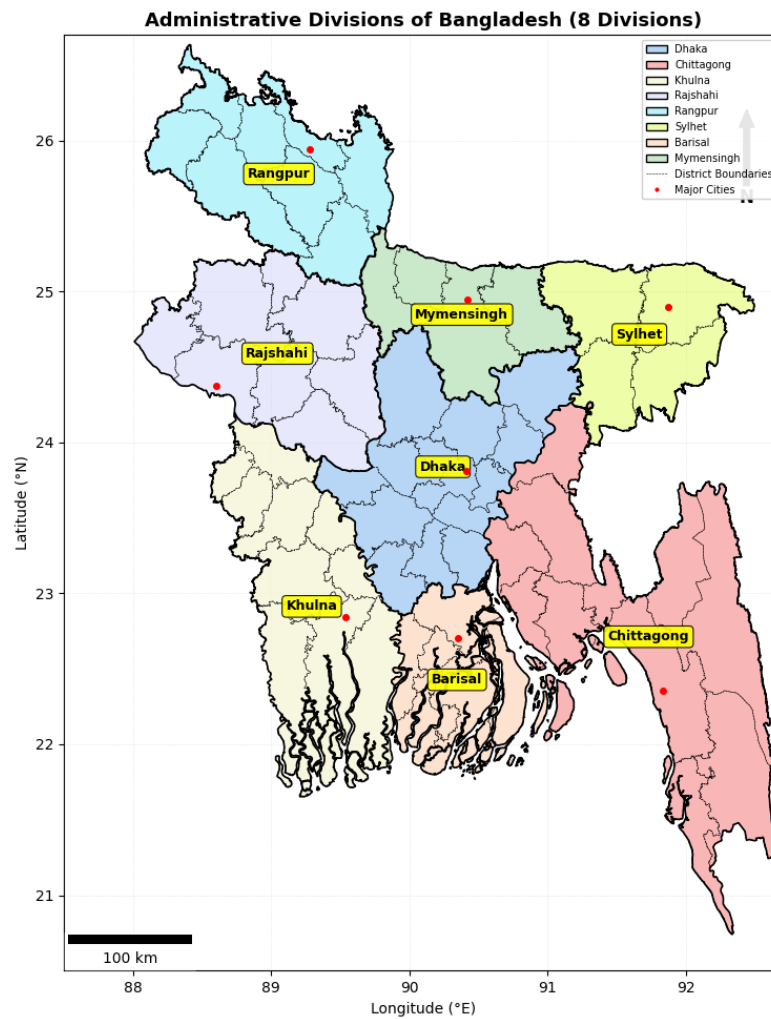


Figure 4.1: Map of the study area

Division	Area (%)	ERA5 grid number	Climate condition
Dhaka	13.96	29	Warm, humid subtropical
Chittagong	22.98	44	Tropical monsoon
Sylhet	8.56	18	Humid subtropical
Rajshahi	12.30	27	Tropical wet and dry
Rangpur	10.97	24	Humid subtropical
Khulna	15.10	28	Tropical monsoon
Barishal	8.97	14	Tropical monsoon
Mymensingh	7.17	13	Humid subtropical

Table 4.1: Description of Bangladesh's eight divisions

4.2.2 Data

4.2.2.1 Data Sources

This study used the ERA5 reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and made available through the Copernicus Climate Data Store (CDS). ERA5 is a state-of-the-art global atmospheric reanalysis that combines numerical weather prediction models with a wide range of observations using data assimilation techniques to provide spatially and temporally consistent climate information [54]. The dataset has a spatial resolution of $0.25^\circ \times 0.25^\circ$ and was downloaded programmatically using the Python CDS API, ensuring efficient and reproducible retrieval of the required meteorological variables. Instead of using station observations from the Bangladesh Meteorological Department (BMD), this study employed the ERA5 hourly data on pressure levels from 1940 to present because it provides complete spatial coverage across Bangladesh and overcomes limitations associated with missing observations, uneven station distribution, and data discontinuities commonly encountered in station-based records. The uniform gridded dataset enables consistent extraction of meteorological variables for each of the eight administrative divisions, making it well suited for long-term climate analysis and machine learning-based prediction of annual HWDs.

4.2.2.2 Study Period

The ERA5 dataset covers the period from 1940 to 2025. For model development, data from 1940–2008 were used for training, while data from 2009 to 2025 were reserved for independent testing and model evaluation.

4.2.2.3 Selection of the climate domain

The ERA5 reanalysis dataset was used to extract atmospheric predictors for the eight administrative divisions of Bangladesh. The selection of an appropriate predictor domain is essential for accurately representing the large-scale atmospheric conditions associated with heat wave occurrence. Bangladesh was divided into its eight administrative divisions, and meteorological variables were extracted from 197 ERA5 grid cells with a spatial resolution of $0.25^\circ \times 0.25^\circ$. The number of grid cells representing each division varies according to its geographical extent, with Chittagong (44 grids) having the largest coverage, followed by Dhaka (29), Rajshahi (27), Sylhet (18), Barishal (14), Khulna (28), Rangpur (24), and Mymensingh (13). Figure 4.2 describes the ERA5 grid selection framework over Bangladesh.

4.2.2.4 Predictor Variables

Atmospheric predictor variables were extracted from the ERA5 hourly pressure-level reanalysis dataset for the pre-monsoon season (March–May), which corresponds to the primary heat-wave season in Bangladesh. Based on previous studies conducted in Bangladesh and the broader South Asian region, seven meteorological variables were selected at four pressure levels (1000, 850, 500, and 300 hPa) to represent the thermal, moisture, circulation, and dynamical characteristics of the atmosphere. Air temperature (t) was selected as the primary predictor because it directly determines the occurrence and intensity of heat waves. Recent analyses of heat waves in Bangladesh using ERA5 reanalysis have demonstrated that anomalously high atmospheric temperatures are the dominant driver of heat-wave events. Atmospheric relative humidity (r) and specific humidity (q) were included because moisture strongly influences surface energy balance, evapotranspiration, and apparent temperature, thereby affecting both the development

ERA5 Spatial Sampling Framework Over Bangladesh

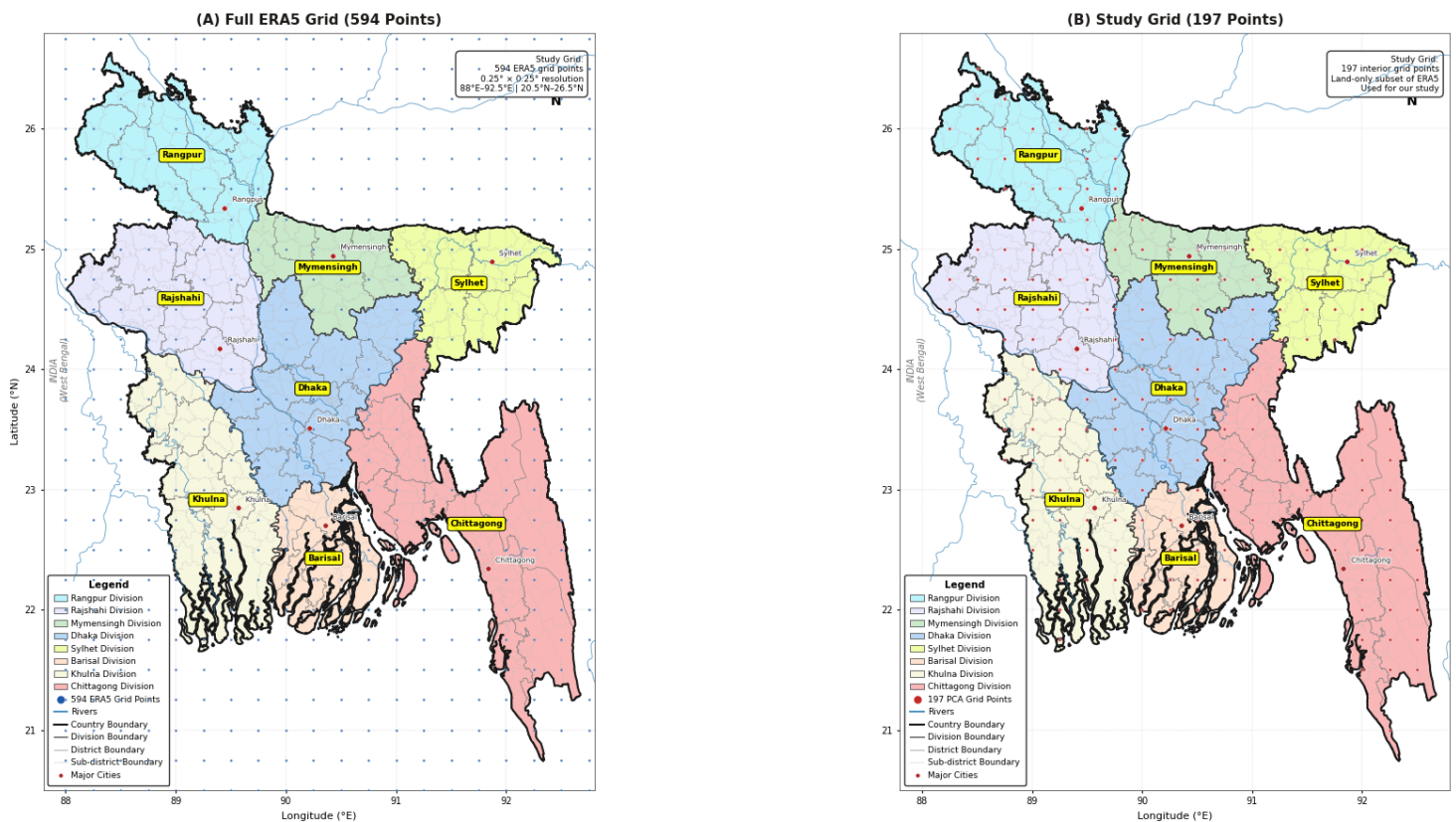


Figure 4.2: ERA5 grid selection framework for Bangladesh. (a) Complete $0.25^\circ \times 0.25^\circ$ ERA5 grid (594 grid points) within the rectangular study domain (88° – 92.5° E, 20.5° – 26.5° N), including both land and surrounding regions. (b) The 197 land-based grid points retained after masking non-land cells, which were used to extract meteorological variables for the eight administrative divisions of Bangladesh

and severity of heat waves. Studies in Bangladesh have shown that changes in atmospheric humidity significantly increase heat stress and amplify extreme temperature conditions. The u-component (u) and v-component (v) of wind were selected to describe horizontal atmospheric circulation and the transport of warm or dry air masses. Persistent circulation anomalies and weak ventilation are frequently associated with prolonged heat-wave episodes across South Asia. Vertical velocity (w) was included to represent large-scale vertical atmospheric motion, where subsidence (downward motion) suppresses cloud formation and precipitation, allowing greater solar heating at the surface. Finally, geopotential (z) was used to characterize the large-scale pressure pattern and upper-air ridges, which are commonly associated with stable atmospheric conditions and persistent heat-wave events. The selected predictor variables and their corresponding ERA5 short names are listed in Table 4.2. By combining thermal, moisture, circulation, and dynamical atmospheric variables from multiple pressure levels, the predictor set captures the key meteorological processes governing heat-wave formation and persistence. Similar atmospheric variables have been successfully employed in previous studies for heat-wave analysis and prediction over Bangladesh and South Asia [55–58].

Table 4.2: Atmospheric variables at different pressure levels

Category	Variable	ERA5 Short Name	Pressure Level (hPa)
Temperature	Temperature	t	1000, 850, 500, 300
Moisture	Relative humidity	r	1000, 850, 500, 300
	Specific humidity	q	1000, 850, 500, 300
Wind/Circulation	U-component of wind	u	1000, 850, 700, 500
	V-component of wind	v	1000, 850, 500, 300
Vertical Motion	Vertical velocity	w	1000, 850, 500, 300
Geopotential	Geopotential	z	1000, 850, 500, 300

4.2.2.5 Definition and estimation of HWDs

There is no universally accepted definition of a heat wave because the characteristics of extreme heat vary across different climatic regions. Consequently, heat waves are commonly defined using either absolute temperature thresholds or percentile-based indices sustained over a minimum number of consecutive days, with the specific criteria depending on the local climate and the objectives of the study [59]. In Bangladesh, the BMD defines a heat wave as a period when the daily maximum temperature exceeds 36°C over a large area and persists for at least three consecutive days [57]. This definition was adopted in the present study because it represents the official operational criterion for heat-wave monitoring in Bangladesh. Since this study employed ERA5 gridded reanalysis data rather than station observations, the BMD definition was adapted to a grid-based framework. For each administrative division, the daily maximum temperature was evaluated for all ERA5 grid cells. A day was classified as a hot day when at least 50% of the grid cells within a division recorded a daily maximum temperature exceeding 36°C. A heat-wave event was then identified when the hot-day condition persisted for three or more consecutive days. HWDs were calculated during the pre-monsoon season (Mar-May), the

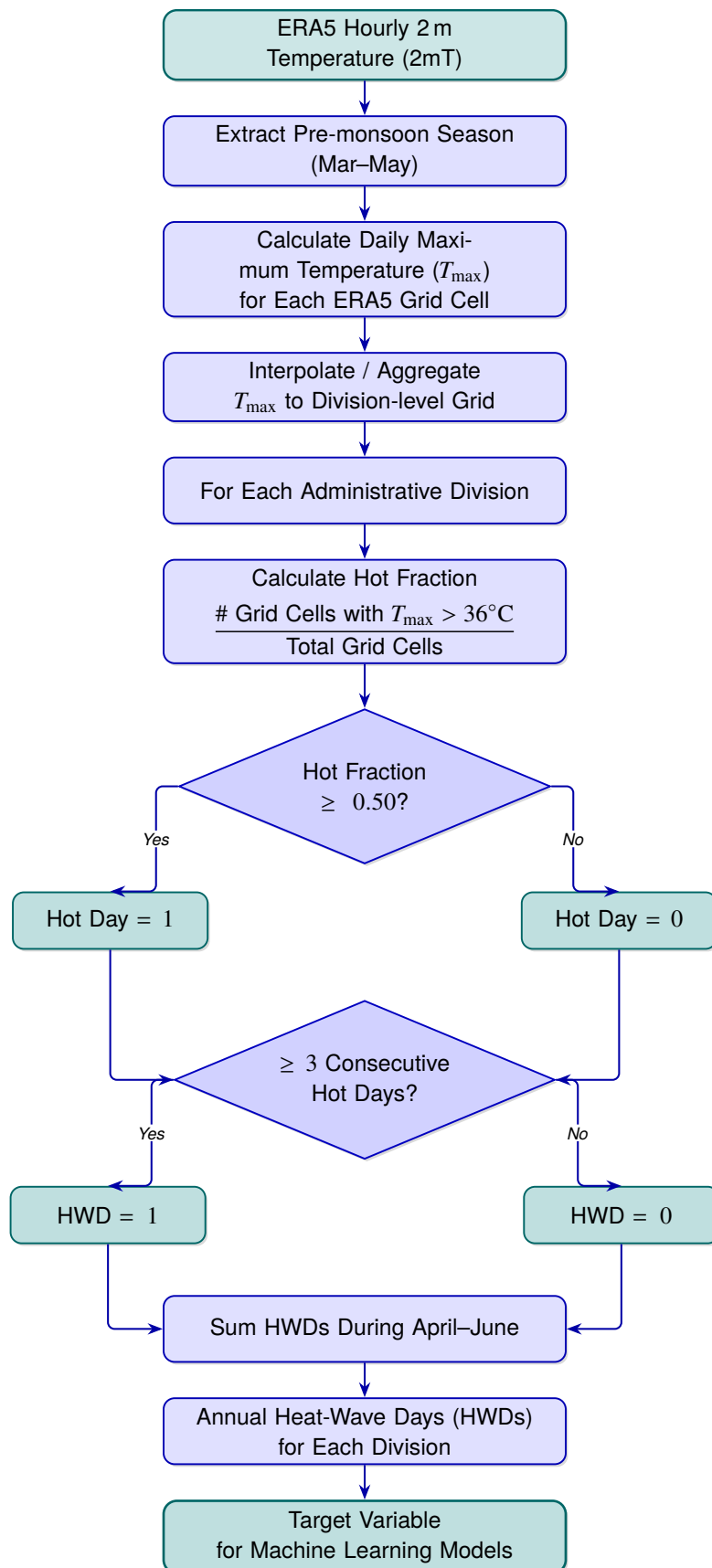


Figure 4.3: Flowchart for calculating annual HWDs from ERA5 hourly 2 m temperature data

period in which heat waves predominantly occur in Bangladesh. The total number of heat-wave days in March, April and May was summed to obtain the annual HWDs for each administrative division. The resulting annual HWDs served as the target variable for developing and evaluating the machine learning models, depicted in the figure 4.3.

4.2.2.6 Dataset Summary

Table 4.3 Summary of the dataset for annual heat wave days prediction in Bangladesh.

Table 4.3: Summary of Study Methodology and Data Parameters

Item	Description
Study area	Eight administrative divisions of Bangladesh
Study period	1940–2025
Predictor period	March–May (Pre-monsoon season)
Target period	Mar–May
Temporal resolution	Annual
Predictor variables	28 (7 variables $\times$ 4 pressure levels)
Atmospheric variables	Temperature, Relative Humidity, Specific Humidity, U-wind, V-wind, Vertical Velocity, Geopotential
Pressure levels	1000, 850, 500, and 300 hPa
Target variable	Annual Heat Wave Days (HWDs)
Feature selection	Minimum Redundancy Maximum Relevance (mRMR)
Feature extraction	Principal Component Analysis (PCA)

4.3. Feature Engineering

Feature engineering was performed to transform the raw ERA5 reanalysis data into meaningful predictor variables suitable for machine learning models. The overall feature engineering workflow, shown in Figure 4.4, consists of three sequential stages: data preprocessing, selection, and feature extraction.

4.3.1 Data Preprocessing

The ERA5 reanalysis data were downloaded using the latitude and longitude coordinates covering the study area. As the downloaded data included grid points outside the geographical boundary of Bangladesh, a Bangladesh national boundary shapefile was used to extract only the ERA5 grid points located within the country. Subsequently, a division-level administrative shapefile was employed to assign each ERA5 grid point to one of the eight administrative divisions of Bangladesh. For each division, the ERA5 dataset contained multiple grid points for every atmospheric variable at four pressure levels (300, 500, 850, and 1000 hPa). To obtain representative annual values for each division, the values from all grid points within a division were averaged for each year and pressure level. This spatial aggregation transformed the grid-level observations into a single annual value for each atmospheric variable within each division. The annual average for each variable was calculated using the following procedure:

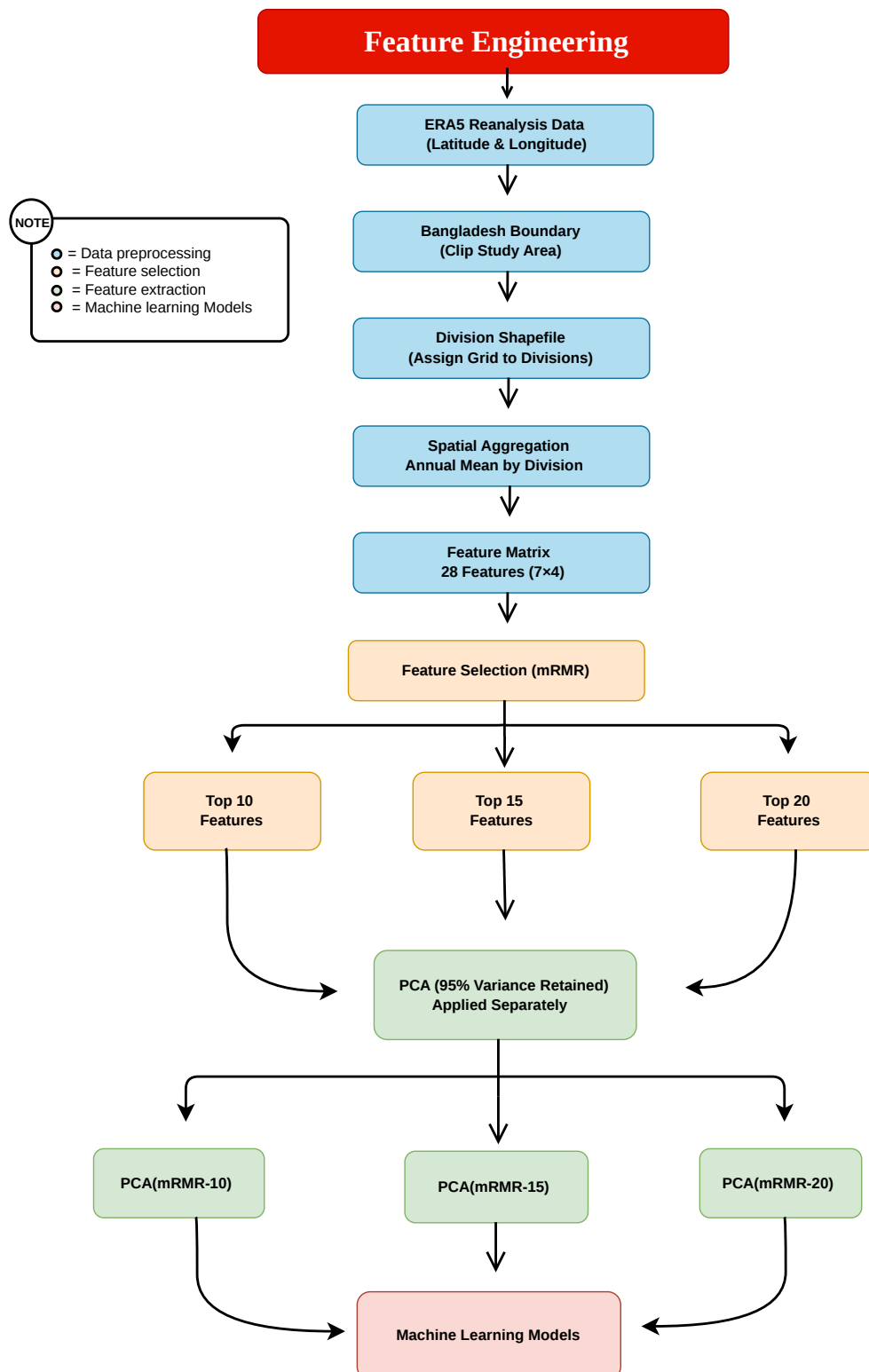


Figure 4.4: feature engineering workflow

```

1 feature_df = (
2     df.groupby(['year', 'pressure_level'])['feature_name']
3     .mean() # mean over all grid points for each division
4     .reset_index()
5 )

```

Listing 4.1: Calculation of annual mean values for each pressure level

The processed data were then reshaped into a matrix format, where each row corresponds to a year and each column represents a pressure level.

```

1 feature_name_matrix = feature_df.pivot_table(
2     index='year',
3     columns=['pressure_level'],
4     values='feature_name'
5 )

```

Listing 4.2: Transformation of the processed data into matrix format

The same preprocessing procedure was applied to all atmospheric predictor variables. Since seven atmospheric variables were considered at four pressure levels, the resulting dataset contained a total of 28 raw predictor features (7 variables $\times$ 4 pressure levels) for each division and year.

4.3.2 Feature Selection

To identify the most relevant predictor variables while reducing redundancy, mRMR algorithm was applied to the 28 raw features. The mRMR algorithm ranks features according to their relevance to the target variable while minimizing redundancy among the selected predictors. Based on the resulting ranking, the top 10, top 15, and top 20 features were selected to create three distinct feature subsets. These subsets were subsequently used as the input for PCA to further reduce their dimensionality.

4.3.3 Feature Extraction

PCA was employed as a feature extraction technique to further reduce the dimensionality of the mRMR-selected feature subsets. PCA was applied separately to each of the three subsets consisting of the top 10, top 15, and top 20 features. For each subset, the number of principal components was determined such that 95% of the total variance was retained. The resulting principal components constituted three transformed feature sets, namely PCA (mRMR-10), PCA (mRMR-15), and PCA (mRMR-20), which were subsequently used as inputs for training and evaluating the machine learning models. Thus, three different feature sets were evaluated in this study:

- **PCA (mRMR-10):** Principal components extracted from the top 10 mRMR-selected features while retaining 95% of the total variance.
- **PCA (mRMR-15):** Principal components extracted from the top 15 mRMR-selected features while retaining 95% of the total variance.

- **PCA (mRMR-20):** Principal components extracted from the top 20 mRMR-selected features while retaining 95% of the total variance.

These three feature sets were independently used to train and evaluate the machine learning models.

4.4. Machine Learning Model Development

Machine learning (ML) is a branch of artificial intelligence that enables computers to learn patterns and relationships directly from data without explicit programming. Owing to advances in computational power, optimization techniques, and the increasing availability of large datasets, ML has become an effective tool for solving complex problems across various scientific disciplines [60]. In climate science, the complex and nonlinear interactions among atmospheric variables make the prediction of extreme events, such as heat waves, particularly challenging using conventional statistical methods. Machine learning provides a data-driven approach capable of capturing these nonlinear relationships, thereby improving predictive performance. Consequently, ML algorithms have been widely applied for forecasting temperature extremes and other climate-related phenomena [61]. To estimate annual HWDs in Bangladesh, six regression-based machine learning models were implemented and compared using the selected atmospheric predictors. The models comprised Linear Regression, Ridge Regression, Lasso Regression, Elastic Net Regression, Gaussian Process Regression (GPR), and Multi-Layer Perceptron (MLP). These algorithms were selected to represent both traditional statistical learning methods and modern neural network-based approaches, allowing a comprehensive comparison of linear, regularized, nonlinear, and deep learning techniques for heat wave prediction.

4.4.1 Overview of Artificial Neural Network

Artificial Neural Networks (ANNs) are computational models inspired by the human brain that learn complex relationships between input and output data. Owing to their ability to model both linear and nonlinear relationships, ANNs are widely used for prediction, classification, and pattern recognition tasks. By learning from historical data, they can generalize underlying patterns and make accurate predictions for unseen observations. Among the various ANN architectures, the MLP is one of the most widely used feed-forward neural networks. It is a supervised learning model that learns the relationship between input and output variables from labelled data [62]. Through iterative optimization of network weights and biases, the MLP constructs a nonlinear mapping that enables accurate prediction of output values, making it well suited for complex environmental and climate prediction problems. The main building blocks of an ANN are:

- **Artificial neurons:** Basic computational units that compute weighted sums of inputs and apply an activation function.
- **Network architecture:** Comprises an input layer, one or more hidden layers, and an output layer.
- **Weights and biases:** Trainable parameters that determine the strength of neuron connections and are updated to minimise prediction error.
- **Training algorithm:** The network learns by minimising a loss function through back-propagation and an optimization algorithm such as gradient descent or Adam [63].

4.4.2 Architecture of MLP

Figure 4.5 illustrates the architecture of MLP. The network consists of four main components: an input layer, two hidden layers, and an output layer. The input layer contains four neurons (X_1 – X_4), representing the predictor variables. Each input neuron is fully connected to every neuron in the first hidden layer, indicating that every feature contributes to the computation of each hidden neuron. For each neuron in the first hidden layer (n_1 – n_5), the incoming inputs are multiplied by their corresponding connection weights and summed together with a bias term. This weighted sum is expressed as

$$z = \sum_{i=1}^n w_i x_i + b, \quad (4.1)$$

where x_i denotes the input features, w_i represents the connection weights, and b is the bias. The weighted sum determines the net input received by each neuron. The computed value z is then transformed by a nonlinear activation function to produce the neuron's output,

$$a = f(z), \quad (4.2)$$

where a is the activated output and $f(\cdot)$ denotes the activation function. In this study, the Rectified Linear Unit (ReLU) activation function is adopted for the hidden layers,

$$f(z) = \max(0, z), \quad (4.3)$$

which enables the network to capture complex nonlinear relationships while mitigating the vanishing gradient problem. The activated outputs from the first hidden layer become the inputs to the second hidden layer (z_1 – z_5), where the same sequence of weighted summation and activation is repeated. Finally, the processed information is propagated to the output layer, consisting of three output neurons (y_1 , y_2 , and y_3). For regression problems, the output layer generally employs a linear activation function to produce the final prediction,

$$\hat{y} = W_o a + b_o, \quad (4.4)$$

where W_o and b_o denote the output-layer weights and bias, respectively. The fully connected architecture allows information from every input variable to influence every neuron in the subsequent layers. During training, the connection weights and biases are iteratively updated using the backpropagation algorithm and an optimization method such as Adam to minimize the prediction error.

4.4.3 Training Hyperparameters

The MLP model was implemented using the Scikit-learn library. The network architecture consisted of a single hidden layer with eight neurons, and the Rectified Linear Unit (ReLU) activation function was employed to model nonlinear relationships between the input features and the target variable. Model parameters were optimized using the Adaptive Moment Estimation (Adam) optimizer with an initial learning rate of 0.001. To improve the generalization capability of the model and reduce overfitting, L2 regularization with a penalty coefficient of 0.001 was applied. The model was trained using a mini-batch size of 8 samples and a maximum of 5000 iterations. Furthermore, early stopping was enabled by reserving 15% of the training data as a validation set. The training process terminated automatically if the validation performance did not improve for 30 consecutive iterations. The configuration of the proposed MLP model is summarized in Table 4.4.

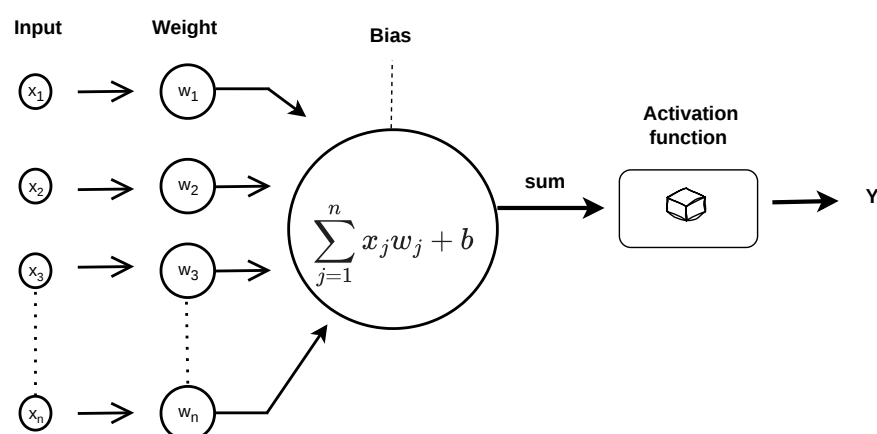
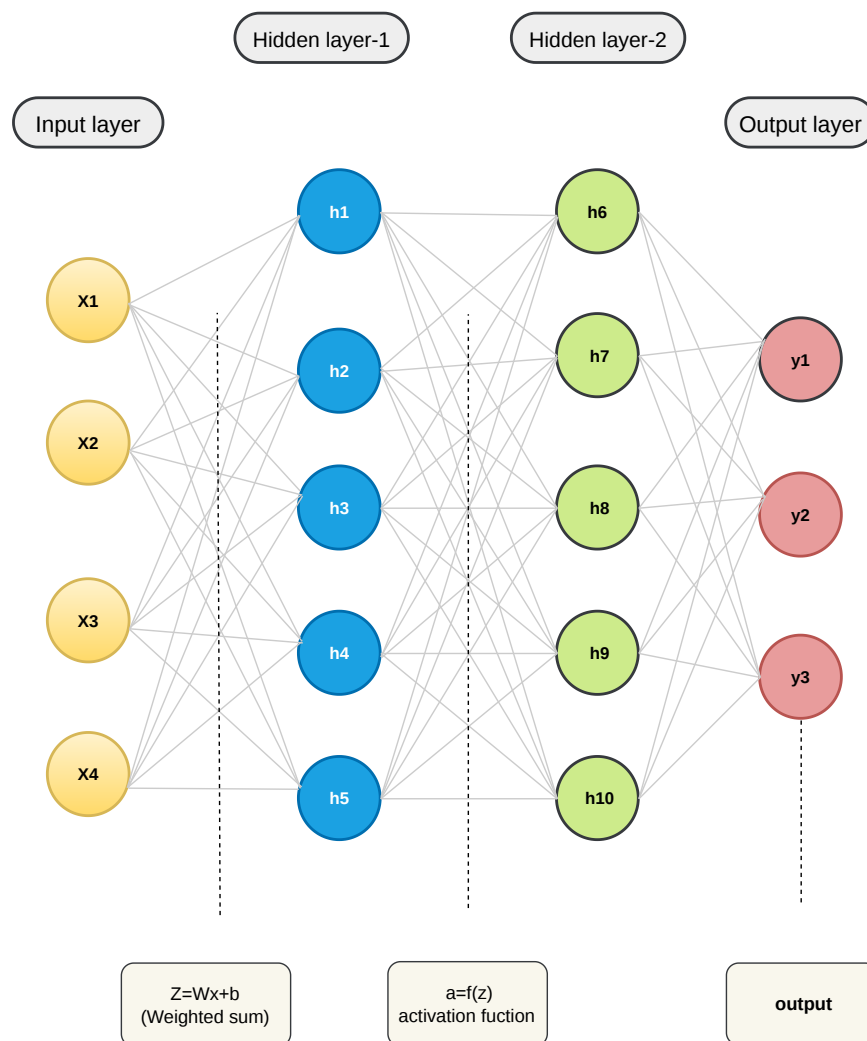


Figure 4.5: Architecture of MLP

Table 4.4: Hyperparameter Settings for the Multi-Layer Perceptron Regressor

Hyperparameter	Value
Model	Multi-Layer Perceptron Regressor
Input features	PCA-transformed features
Hidden layers	1
Neurons	8
Activation function	ReLU
Output activation	Linear (default)
Optimizer	Adam
Initial learning rate	0.001
Learning rate schedule	Adaptive
L2 regularization (α)	0.001
Batch size	8
Maximum iterations	5000
Early stopping	Enabled
Validation fraction	15%
Patience	30

4.4.4 Model Evaluation Metrics

1. Mean Absolute Error (MAE)

MAE measures the average magnitude of the errors in a set of predictions, without considering their direction. It is the average absolute difference between the predicted values and actual values.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4.5)$$

Where:

- n is the total number of observations.
- y_i is the actual value for the i -th observation.
- $\hat{y}_i$ is the predicted value for the i -th observation.

2. Root Mean Squared Error (RMSE)

RMSE is a quadratic scoring rule that measures the average magnitude of the error. It gives a relatively high weight to large errors, making it useful when large errors are particularly undesirable.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4.6)$$

3. Correlation Coefficient (CC)

The Correlation Coefficient (often Pearson's r) measures the strength and direction of the linear relationship between the actual and predicted values.

$$CC = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}} \quad (4.7)$$

Where:

- $\bar{\hat{y}}$ is the mean of the predicted values.

4.5. Overall Workflow

Figure 4.6 illustrates the overall workflow adopted in this study for predicting annual HWDs in Bangladesh. The methodology begins with the collection of ERA5 reanalysis data, including hourly 2 m air temperature and pressure-level atmospheric variables, followed by data pre-processing to extract Bangladesh-specific meteorological information and generate predictor and target datasets. The processed data are then divided into training (1940–2008) and testing (2009–2025) sets, after which feature engineering techniques, including standardization, mRMR feature selection, and PCA, are applied. Finally, multiple machine learning models are developed and evaluated using MAE, RMSE and CC, and their performances are compared to identify the most effective model for annual heat-wave prediction.

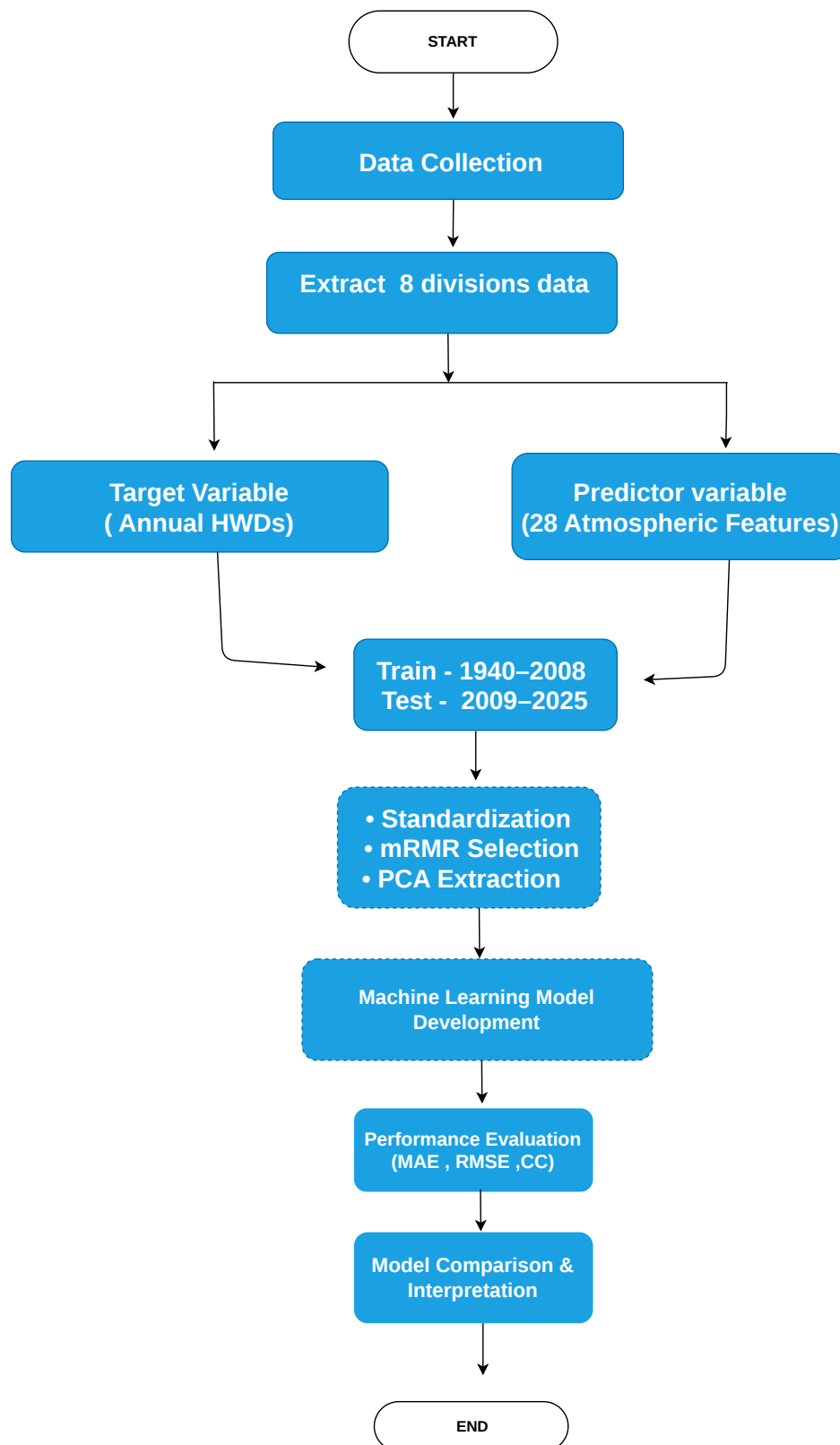


Figure 4.6: Overall methodological workflow

4.6. Results and Discussion

4.6.1 Heat Wave Characteristics in Bangladesh

4.6.1.1 Spatial Distribution of Heat Wave Days

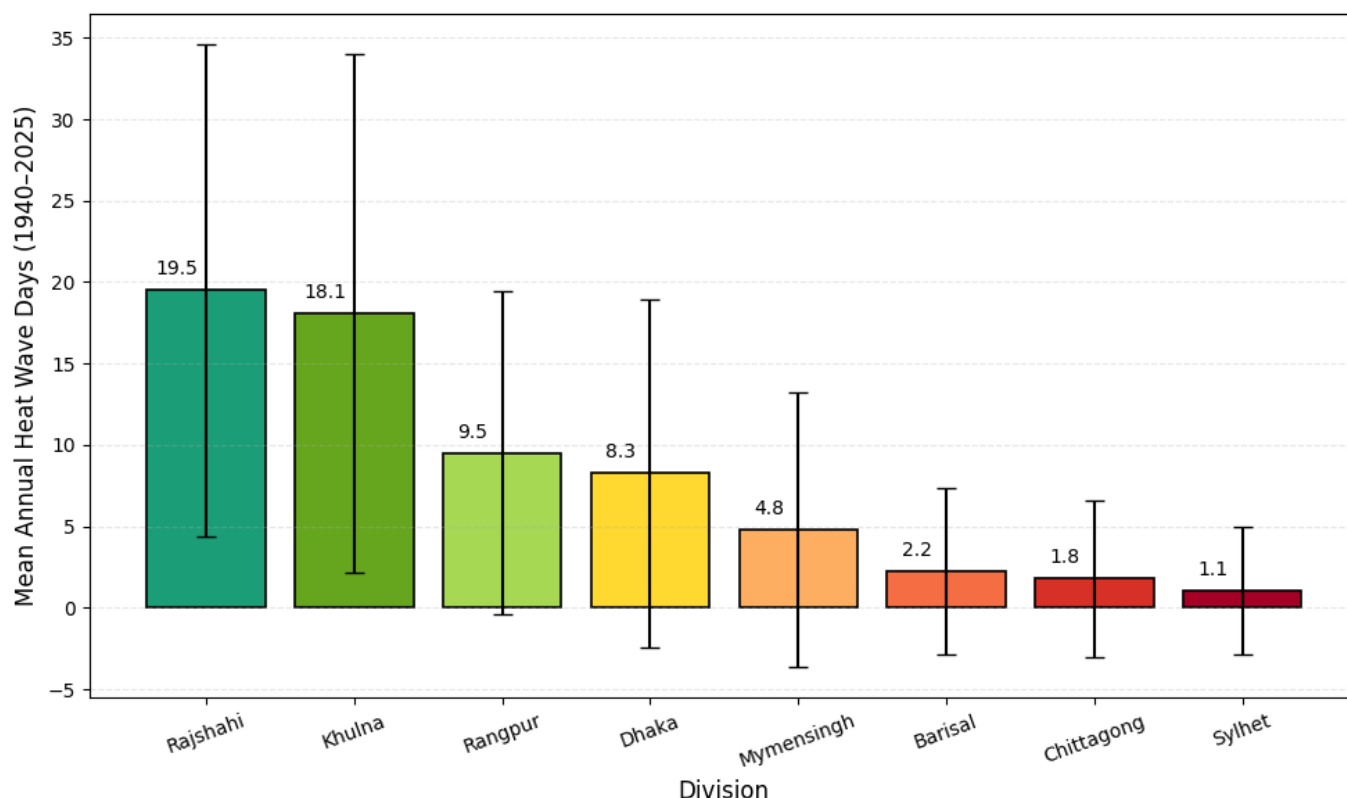


Figure 4.7: Spatial distribution of average annual HWDs across bangladesh's divisions

The spatial distribution of annual HWDs across Bangladesh exhibits a pronounced west-to-east gradient (Figure 4.7 and Table 4.5). Rajshahi and Khulna experience the highest mean annual HWDs, averaging 19.49 and 18.08 days, respectively, whereas Sylhet (1.06 days), Chittagong (1.80 days), and Barisal (2.24 days) record the lowest values. Rangpur (9.50 days), Dhaka (8.27 days), and Mymensingh (4.81 days) show intermediate heat wave activity. The interpolated spatial pattern shown in Figure 4.8 further illustrates a continuous decrease in heat wave intensity from the western regions toward the eastern and southeastern parts of Bangladesh. To visualize the spatial variability of heat wave occurrence, heat wave days (HWDs) were first calculated for each ERA5 grid cell using a grid-based definition, whereby a heat wave was defined as three or more consecutive days with daily maximum temperature (T_{max}) greater than or equal to 36°C at an individual grid cell. The resulting grid-based HWD values were subsequently interpolated to generate a continuous spatial surface. The highest heat wave occurrence is concentrated over Rajshahi and northern Khulna, while the lowest values occur in Sylhet, Chittagong, and the coastal regions. This west-to-east gradient is consistent with Bangladesh's climatology, where the drier continental climate of the northwest favours more frequent and persistent heat waves, whereas the eastern and coastal regions experience lower heat wave occurrence due to the moderating influence of the Bay of Bengal and enhanced moisture transport from the surrounding hill ranges.

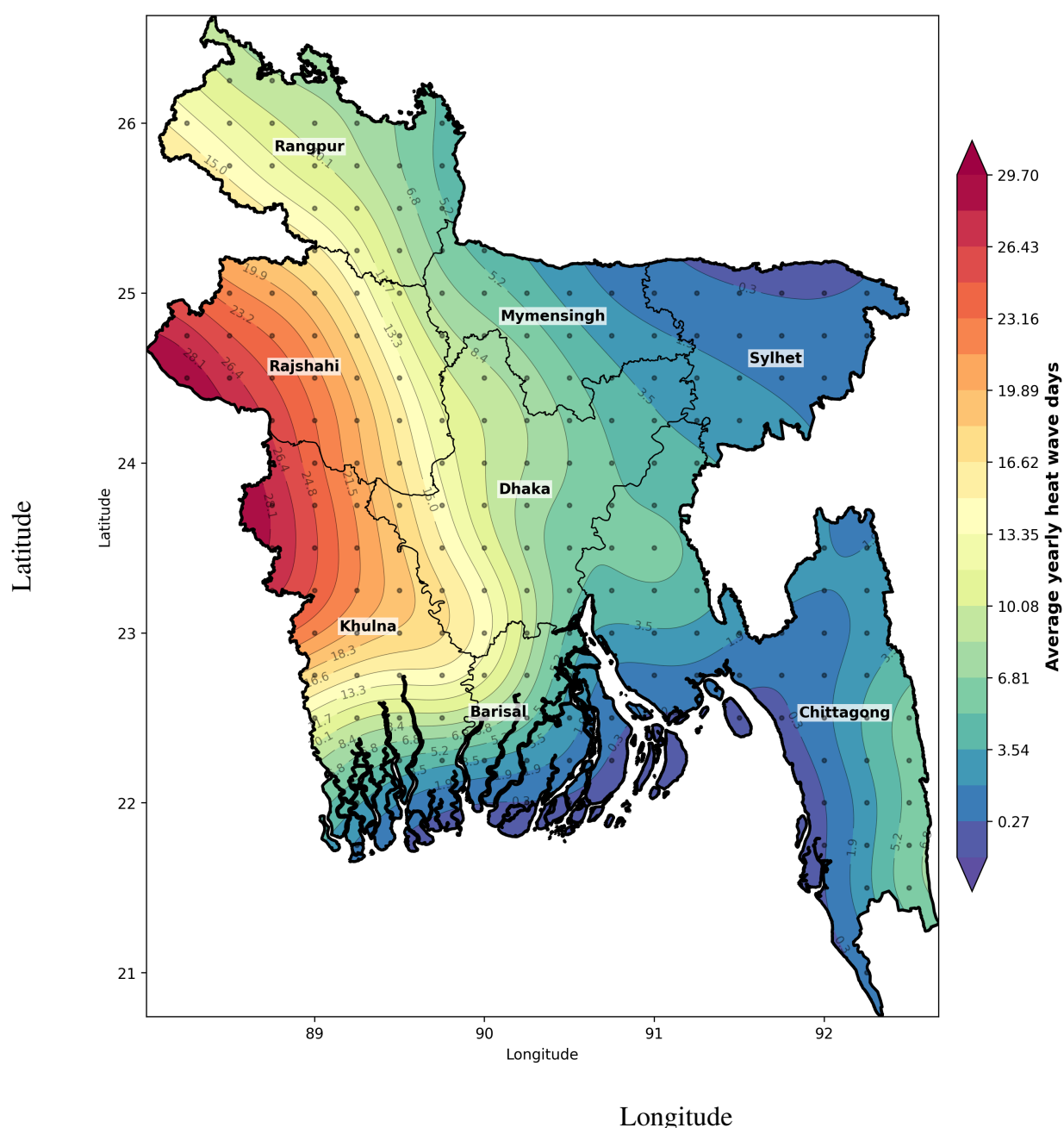


Figure 4.8: Spatial pattern of heat wave occurrence in Bangladesh (1940–2025)

4.6.1.2 Temporal Variation of Heat Wave Days

Figure 4.9 illustrates the annual HWD time series for all eight divisions during 1940–2025. A prominent nationwide peak occurred in 1957, representing the highest recorded HWDs in seven of the eight divisions, indicating a large-scale regional heat wave event. Rajshahi and Khulna consistently experienced the highest heat wave activity throughout the study period, although both exhibited a declining tendency after the 1960s followed by a modest recovery in recent decades. In contrast, Mymensingh, Barisal, and Sylhet recorded very few heat wave days during the 1990s–2000s before showing slight increases in the last decade. Chittagong displayed a unique pattern, with relatively low historical HWDs but a noticeable increase after 2020, reaching its maximum in 2023. The long-term linear trend analysis presented in Table 4.5 indicates that only Khulna ($-1.44 \text{ days decade}^{-1}$, $p = 0.036$) and Mymensingh

(-0.75 days decade $^{-1}$, $p = 0.039$) exhibit statistically significant decreasing trends. Rajshahi also shows a declining trend, although it is not statistically significant, while Rangpur and Chittagong exhibit weak positive but non-significant trends. Overall, no division demonstrates a statistically significant increase in annual HWDs during 1940–2025, suggesting substantial interannual variability rather than a consistent long-term increase in heat wave occurrence across Bangladesh.

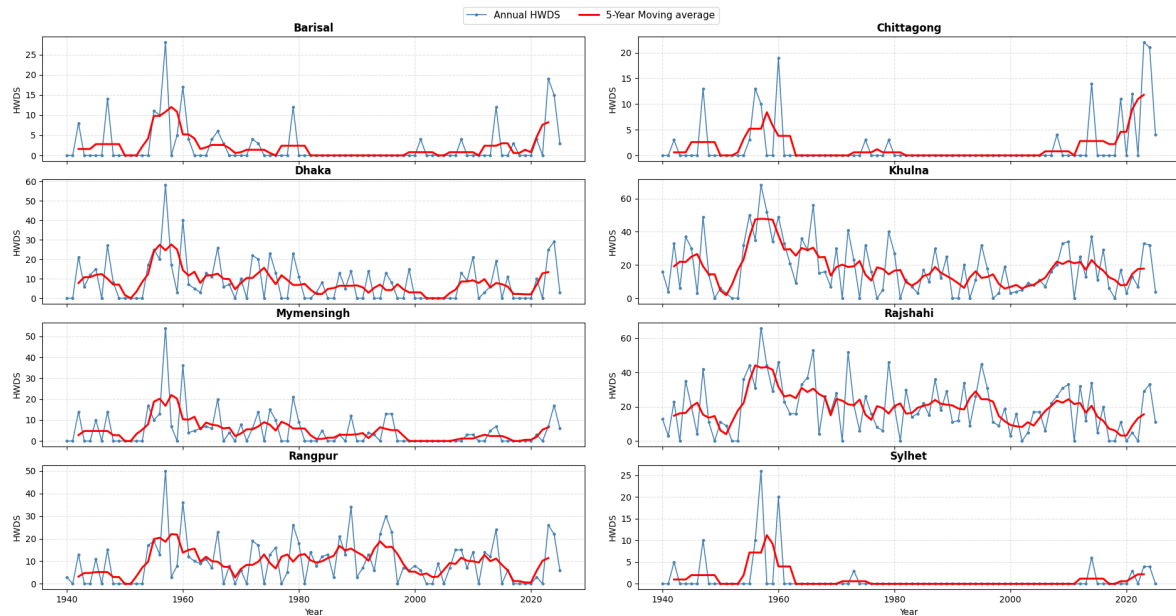


Figure 4.9: Annual HWDs by division, 1940–2025, with 5-year moving average (red)

4.6.1.3 Division-wise Comparison of Heat Wave Days

Figure 4.10 compare the annual HWD distributions across the eight administrative divisions during 1940–2025.

Table 4.5: Summary statistics of annual HWDs by division, 1940–2025. Trend p-values marked * are statistically significant at the 0.05 level.

Division	Mean HWD	Std. Dev.	Minimum	Maximum	Range	Trend (days)	p-value
Rajshahi	19.49	15.08	0	66	0–66	-1.02	0.12
Khulna	18.08	15.92	0	68	0–68	-1.44	0.036*
Rangpur	9.50	9.92	0	50	0–50	0.06	0.893
Dhaka	8.27	10.69	0	58	0–58	-0.80	0.085
Mymensingh	4.81	8.44	0	54	0–54	-0.75	0.039*
Barisal	2.24	5.07	0	28	0–28	-0.13	0.56
Chittagong	1.80	4.82	0	22	0–22	0.30	0.147
Sylhet	1.06	3.90	0	26	0–26	-0.25	0.139

Rajshahi and Khulna exhibit the highest mean and median HWDs. In contrast, Sylhet, Chittagong, and Barisal show low median HWDs, narrow distributions, and smaller maximum values, reflecting relatively infrequent and less intense heat wave occurrences. Dhaka and Rangpur occupy an intermediate position, with moderate average HWDs but relatively large variability, resulting in occasional extreme years consistent with the nationwide 1957 heat wave

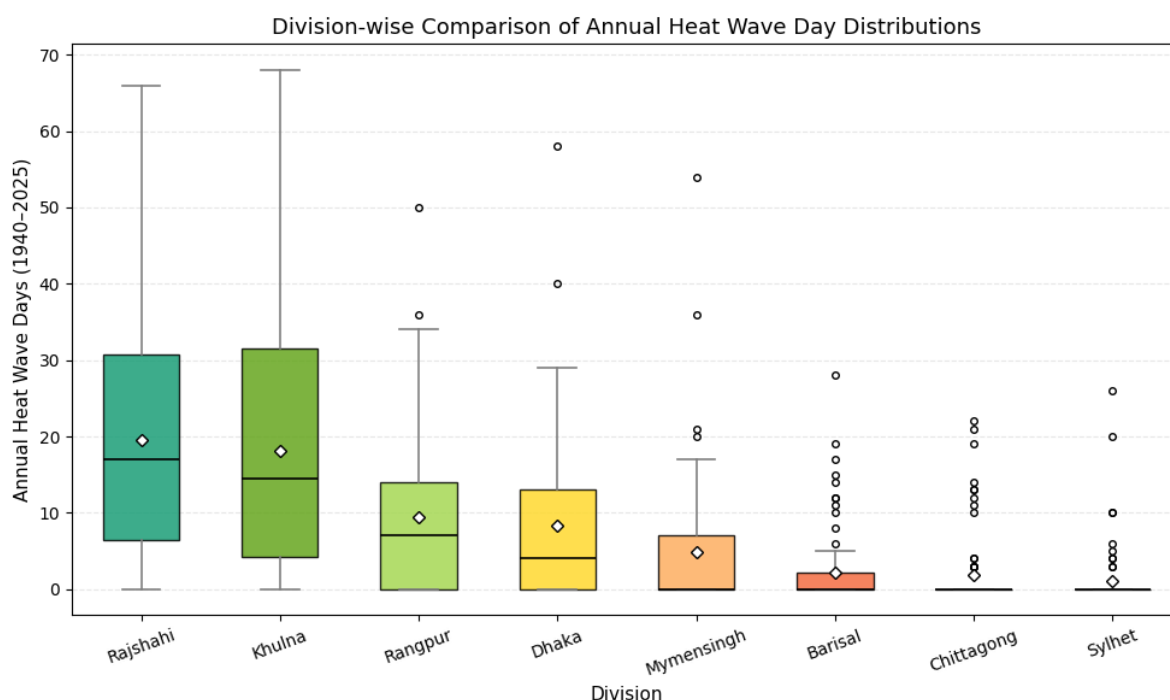


Figure 4.10: Box plots of annual heat wave day distributions by division (1940–2025). Boxes show interquartile range and median; diamonds show the mean; whiskers extend to $1.5 \times \text{IQR}$; open circles denote outliers.

event shown in Figure 4.9. Overall, the results presented in Figures 4.7 - 4.10 and Table 4.5 demonstrate that heat wave activity in Bangladesh is concentrated in the western divisions, particularly Rajshahi and Khulna, while the eastern and coastal regions remain comparatively less affected.

4.6.2 Machine Learning Model Performance

The performance comparison of the six machine learning models using PCA features derived from the mRMR-selected features demonstrates that prediction accuracy varies across the eight divisions of Bangladesh and the three PCA configurations (Tables 4.6–4.8). Overall, the four linear regression-based models (Linear Regression, Ridge Regression, Lasso Regression, and Elastic Net Regression) produced highly consistent results across all divisions, with only minor differences in MAE, RMSE, and CC. Among these models, Ridge Regression generally achieved slightly lower prediction errors while maintaining stable correlation coefficients, indicating that L2 regularization provided marginal improvements in model generalization. The Gaussian Process GPR model exhibited mixed performance across the study area. Although it achieved competitive results in several divisions, particularly Sylhet and Mymensingh, its prediction accuracy deteriorated in other regions, resulting in relatively larger errors and lower, or occasionally negative, correlation coefficients. Similarly, the MLP model showed substantial variability in performance. The MLP achieved the lowest prediction errors in Mymensingh using PCA (mRMR-10) and in Sylhet using PCA (mRMR-20), demonstrating its capability to capture complex nonlinear relationships. However, its performance was inconsistent across the remaining divisions, particularly in Rajshahi and Khulna, suggesting that the limited number of annual observations was insufficient for stable neural network training. A comparison of the

three PCA configurations indicates that PCA features derived from the mRMR-20 selected features generally provided superior predictive performance for most divisions. This configuration yielded lower MAE and RMSE values for the linear regression models in Dhaka, Chittagong, Khulna, Rajshahi, Rangpur, and Sylhet, indicating that retaining twenty informative predictors preserved more useful climatic information than using ten or fifteen features. In contrast, the PCA (mRMR-10) and PCA (mRMR-15) configurations showed comparatively higher prediction errors in several regions. The prediction performance also varied considerably among the eight divisions. Mymensingh consistently recorded the lowest MAE and RMSE values across most models, indicating that annual HWDs in this division were comparatively easier to predict. Rajshahi also exhibited high correlation coefficients ($CC > 0.90$), reflecting strong agreement between observed and predicted HWDs. Conversely, Khulna consistently produced the largest prediction errors, suggesting greater spatial and temporal complexity in its heat wave characteristics. Although Sylhet generally exhibited relatively low MAE and RMSE values, its correlation coefficients were lower than those observed in Rajshahi and Mymensingh, indicating weaker temporal agreement despite small absolute prediction errors. Overall, the results suggest that the linear regression-based models, particularly Ridge Regression, provide the most robust and consistent performance for predicting annual heat wave days across Bangladesh. While Gaussian Process Regression and Multi-Layer Perceptron achieved superior performance in a few individual divisions, their predictive capability was less stable across the entire study area. Furthermore, the PCA features derived from the mRMR-20 selected features consistently provided the best balance between dimensionality reduction and predictive accuracy, making this configuration the most suitable for regional HWD prediction in Bangladesh.

Table 4.6: Performance of ML models for HWD prediction using PCA (mRMR-10)

Model	Metric	Barisal	Chittagong	Dhaka	Khulna	Mymensingh	Rajshahi	Rangpur	Sylhet
Linear Regression	MAE	3.5570	5.1772	4.5262	8.3431	2.0715	6.5215	4.7658	1.5468
	RMSE	5.9716	8.3331	6.7867	10.4511	2.4475	7.8535	6.6624	1.8523
	CC	0.3248	0.5464	0.8476	0.8255	0.8844	0.9066	0.7822	0.5452
Ridge Regression	MAE	3.5578	5.1662	4.5325	8.3415	2.0431	6.5463	4.7230	1.5250
	RMSE	5.9782	8.3265	6.8023	10.4575	2.4287	7.8660	6.6333	1.8314
	CC	0.3218	0.5484	0.8470	0.8267	0.8837	0.9066	0.7819	0.5493
Lasso Regression	MAE	3.5658	5.1760	4.5291	8.3371	2.0660	6.5187	4.7600	1.5512
	RMSE	5.9863	8.3369	6.7947	10.4502	2.4443	7.8429	6.6607	1.8524
	CC	0.3195	0.5441	0.8472	0.8261	0.8843	0.9066	0.7820	0.5437
Elastic Net Reg.	MAE	3.5615	5.1727	4.5299	8.3395	2.0589	6.5288	4.7479	1.5414
	RMSE	5.9812	8.3327	6.7961	10.4529	2.4393	7.8525	6.6513	1.8449
	CC	0.3211	0.5460	0.8472	0.8262	0.8841	0.9066	0.7820	0.5459
Gaussian Process	MAE	3.7145	5.0163	5.0905	8.6877	1.9521	7.0838	4.6231	1.1571
	RMSE	6.6979	8.7704	8.0887	10.9004	2.3996	8.1363	7.4395	1.7026
	CC	0.1505	0.2805	0.7363	0.8065	0.8436	0.8915	0.6807	0.7126
MLP	MAE	3.3994	4.7768	5.3229	8.3377	1.4376	8.9699	4.6206	1.7001
	RMSE	6.1239	8.5222	8.1734	10.2426	1.8693	10.7618	7.3266	2.1537
	CC	0.2584	0.8075	0.7605	0.7904	0.9038	0.7435	0.7442	0.3109

4.6.2.1 Spatial Distribution of Model Error: MAE Heatmaps

Figures 4.11 present the MAE heatmaps of the six machine learning models using PCA-transformed mRMR-selected features across the eight administrative divisions. The heatmaps

Table 4.7: Performance of ML models for HWD prediction using PCA (mRMR-15)

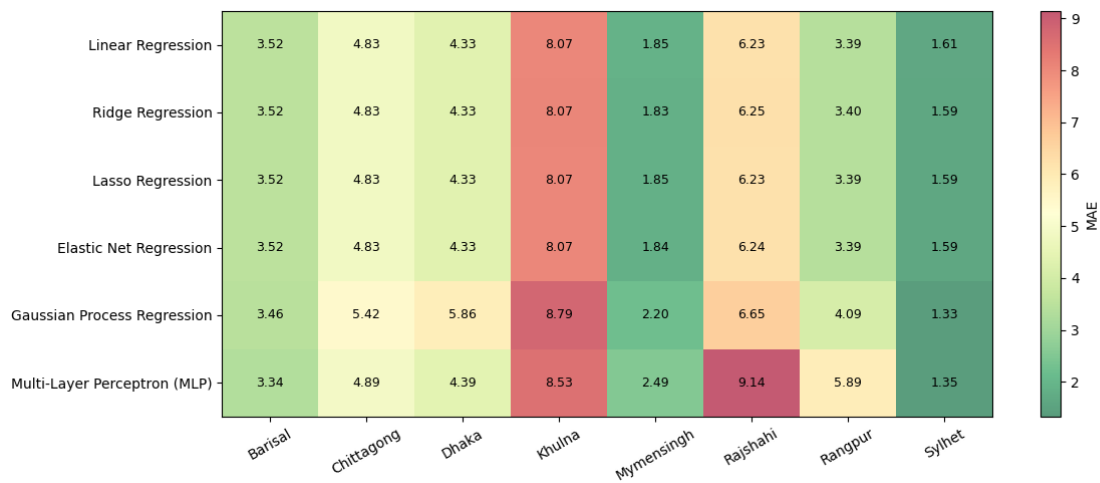
Model	Metric	Barisal	Chittagong	Dhaka	Khulna	Mymensingh	Rajshahi	Rangpur	Sylhet
Linear Regression	MAE	3.5241	4.8323	4.3270	8.0697	1.8511	6.2295	3.3897	1.6099
	RMSE	5.6875	7.9590	6.2693	9.7544	2.4052	7.7017	5.3317	1.9057
	CC	0.5885	0.6178	0.8490	0.8336	0.8846	0.9005	0.8262	0.4822
Ridge Regression	MAE	3.5182	4.8299	4.3308	8.0749	1.8324	6.2532	3.3983	1.5882
	RMSE	5.7005	7.9652	6.2919	9.7890	2.3697	7.7170	5.3421	1.8783
	CC	0.5849	0.6191	0.8497	0.8344	0.8849	0.9010	0.8262	0.4927
Lasso Regression	MAE	3.5163	4.8303	4.3297	8.0692	1.8497	6.2253	3.3858	1.5927
	RMSE	5.6945	7.9671	6.2840	9.7660	2.3957	7.6913	5.3313	1.8854
	CC	0.5852	0.6173	0.8492	0.8342	0.8843	0.9008	0.8259	0.4903
Elastic Net Reg.	MAE	3.5182	4.8304	4.3296	8.0713	1.8439	6.2357	3.3867	1.5937
	RMSE	5.6955	7.9652	6.2845	9.7723	2.3878	7.7019	5.3333	1.8859
	CC	0.5856	0.6180	0.8493	0.8342	0.8846	0.9008	0.8260	0.4899
Gaussian Process	MAE	3.4583	5.4206	5.8568	8.7906	2.1964	6.6492	4.0922	1.3312
	RMSE	6.3894	9.1755	8.7392	10.9538	2.8071	7.9531	6.3042	2.0054
	CC	0.5449	-0.3183	0.6581	0.8096	0.7727	0.8875	0.7631	0.3353
MLP	MAE	3.3381	4.8896	4.3924	8.5335	2.4918	9.1418	5.8864	1.3499
	RMSE	5.9701	8.2041	6.9326	10.1374	3.1454	11.0090	7.8675	1.9656
	CC	0.4962	0.3915	0.9246	0.7781	0.7081	0.8110	0.4802	0.3602

Table 4.8: Performance of ML models for HWD prediction using PCA (mRMR-20)

Model	Metric	Barisal	Chittagong	Dhaka	Khulna	Mymensingh	Rajshahi	Rangpur	Sylhet
Linear Regression	MAE	3.8818	4.6823	3.6741	6.6121	2.5390	6.1863	3.6994	1.4869
	RMSE	6.0266	7.8574	5.6321	7.7093	3.3758	7.5831	5.1301	1.7939
	CC	-0.5036	0.6772	0.8675	0.8274	0.8712	0.8929	0.8327	0.5604
Ridge Regression	MAE	3.8720	4.6802	3.6939	6.6107	2.4774	6.1868	3.6767	1.4734
	RMSE	6.0305	7.8573	5.6572	7.7402	3.2728	7.5786	5.1236	1.7752
	CC	0.5018	0.6784	0.8678	0.8283	0.8720	0.8936	0.8346	0.5672
Lasso Regression	MAE	3.8708	4.6858	3.6845	6.6109	2.5257	6.1900	3.7012	1.4728
	RMSE	6.0329	7.8617	5.6486	7.7179	3.3465	7.5857	5.1312	1.7779
	CC	0.5014	0.6762	0.8675	0.8277	0.8714	0.8931	0.8329	0.5646
Elastic Net Reg.	MAE	3.8729	4.6833	3.6862	6.6111	2.5091	6.1884	3.6922	1.4751
	RMSE	6.0311	7.8595	5.6490	7.7242	3.3245	7.5828	5.1281	1.7794
	CC	-0.5019	0.6771	0.8676	0.8279	0.8716	0.8933	0.8335	0.5649
Gaussian Process	MAE	3.5184	5.5582	5.1174	7.2500	2.0022	6.2464	3.7379	1.4281
	RMSE	6.4832	9.3696	8.0391	8.8594	2.5073	7.5124	5.7696	2.2200
	CC	-0.3108	-0.3350	0.7376	0.8033	0.8184	0.8973	0.8177	0.0416
MLP	MAE	5.2967	5.3518	3.9048	7.2951	2.3990	8.6101	5.6421	1.1885
	RMSE	8.5575	8.8584	6.0874	8.9093	2.8478	9.9608	7.2171	1.9620
	CC	-0.4707	0.3985	0.8630	0.7419	0.7694	0.7693	0.5912	0.4064



(a) PCA(mRMR-10)



(b) PCA(mRMR-15)



(c) PCA(mRMR-20)

Figure 4.11: Comparison of MAE heatmaps for PCA-transformed features selected using mRMR with Top-10, Top-15, and Top-20 features across the eight divisions of Bangladesh.

Table 4.9: Division-wise Summary of Top Contributing Features and SHAP Influence for Heat Wave Day (HWD) Prediction

Division	Top Contributing Features	SHAP Influence
Barisal	shum_P1000, temp_P850, shum_P850	Moderate to high
Chittagong	shum_P1000, shum_P850, shum_P500	Moderate
Dhaka	shum_P1000, temp_P850, zpt_P1000	Very high (≈ -11 to $+8$)
Khulna	shum_P1000, zpt_P1000, temp_P850	Highest (≈ -12 to $+12$)
Mymensingh	shum_P1000, temp_P850, uwnd_P1000	High
Rajshahi	zpt_P850, temp_P300, temp_P850	High
Rangpur	shum_P1000, temp_P850, temp_P300	High
Sylhet	vvel_P850, temp_P850, shum_P1000	Moderate

show that the four linear models (Linear Regression, Ridge, Lasso, and Elastic Net) exhibit nearly identical prediction errors, indicating that PCA effectively reduced multicollinearity among the selected features. A clear spatial pattern is observed, with Sylhet and Mymensingh consistently achieving the lowest MAE, whereas Rajshahi and Khulna exhibit the highest prediction errors, reflecting greater variability in heat wave characteristics within these divisions. Among the evaluated models, Gaussian Process Regression and the Multi-Layer Perceptron show comparatively higher and more variable errors, particularly in Rajshahi and Khulna, while the linear models provide the most stable performance across all feature sets. Increasing the number of mRMR-selected features from Top-10 to Top-20 generally improves prediction accuracy, with the average MAE decreasing from 4.57 (Top-10) to 4.23 (Top-15) and 4.10 (Top-20). Consequently, the PCA(mRMR-20) feature set provided the best overall balance between predictive accuracy and model efficiency across all divisions. To identify the most suitable model for subsequent interpretation, the average MAE across all eight divisions was computed for each of the six regression models evaluated on this feature set. Ridge Regression achieved the lowest average MAE (3.58), narrowly outperforming Elastic Net Regression (3.59), Linear Regression, and Lasso Regression (both approximately 3.59–3.60). Given the near-identical performance of these four linear models, this marginal advantage is attributed to the mild regularization effect of Ridge Regression rather than any substantial difference in predictive capability, indicating that a linear model is sufficient to capture the dominant structure in the reduced feature space. Accordingly, Ridge Regression was selected as the representative model for the remainder of this study. Although PCA was employed to improve predictive performance, SHAP analysis was performed using the Top-20 mRMR-selected original meteorological variables rather than the PCA-transformed features. This approach enables direct physical interpretation of the contribution of each meteorological variable to the predicted Heat Wave Days, while preserving the predictive insights obtained from the PCA-based feature selection process.

4.6.3 Model Interpretation Using SHAP

Although machine learning models can achieve high predictive accuracy, they often provide limited insight into how individual input variables influence model predictions. To address this

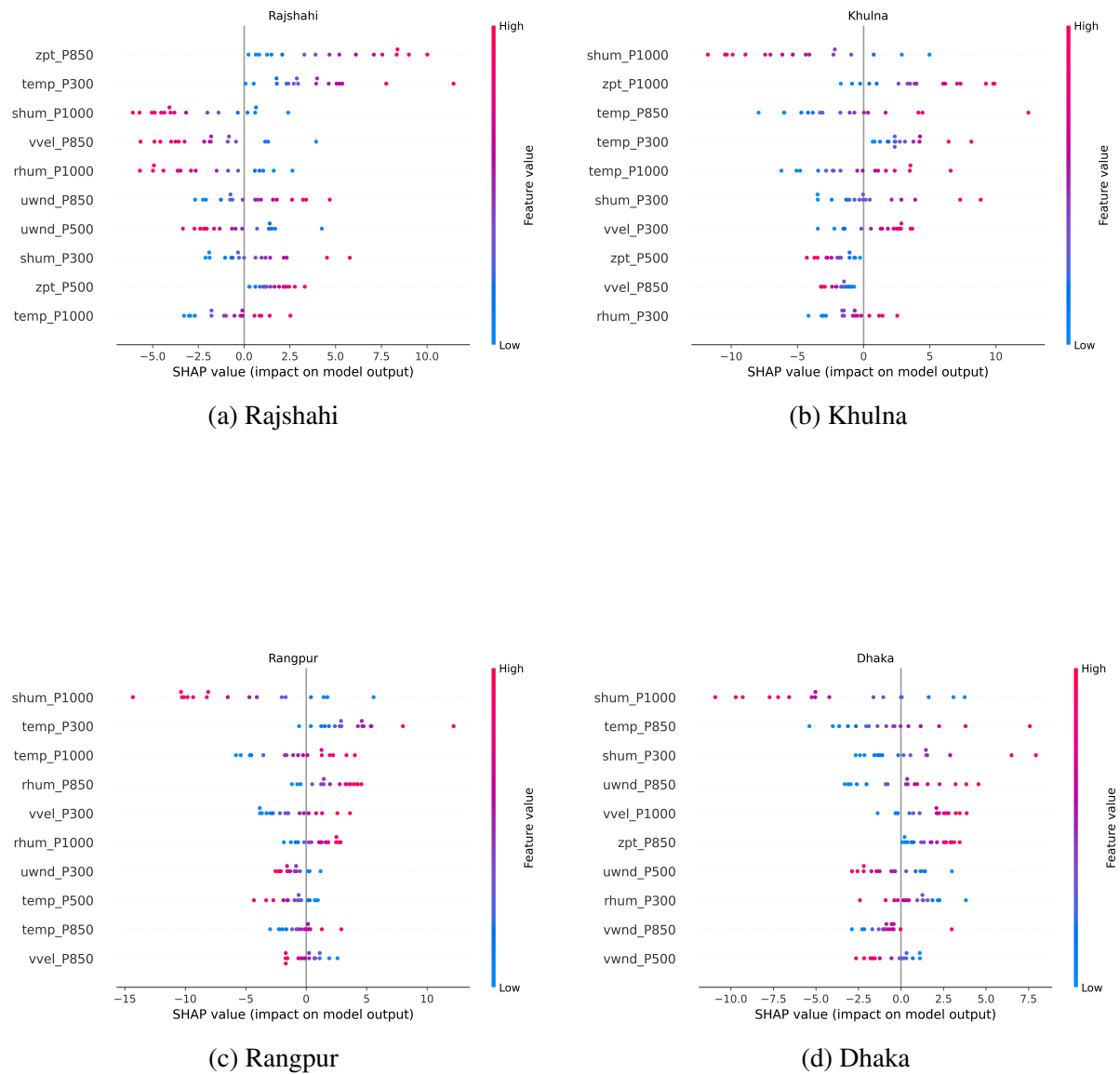


Figure 4.12: SHAP summary (beeswarm) plots of the Ridge Regression model for Rajshahi, Khulna, Rangpur, and Dhaka using the Top-20 mRMR-selected features

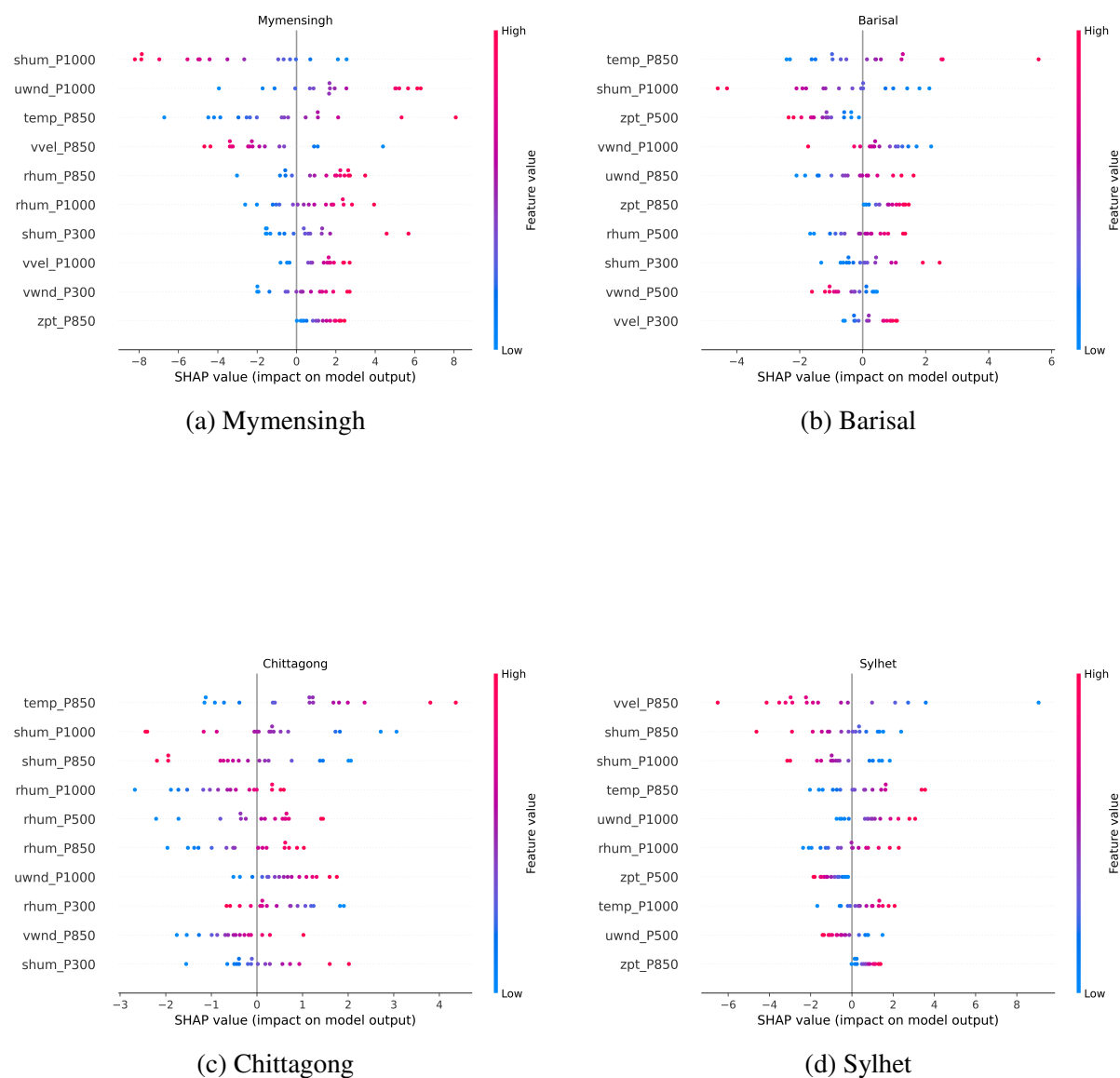


Figure 4.13: SHAP summary (beeswarm) plots of the Ridge Regression model for mymensingh, barisal, Chittagong, Sylhet using the Top-20 mRMR-selected features

limitation, Explainable Artificial Intelligence (XAI) techniques have been developed to improve the transparency and interpretability of predictive models. Among these methods, SHAP has emerged as one of the most widely adopted approaches because it offers both local and global explanations while maintaining a strong theoretical foundation based on cooperative game theory. SHAP was introduced by Lundberg and Lee (2017) as a unified framework for interpreting the predictions of machine learning models by assigning each input feature a contribution value, known as the Shapley value, that quantifies its influence on an individual prediction.[64] Unlike conventional feature importance measures, SHAP satisfies three desirable properties: local accuracy, consistency, and missingness, ensuring that feature attributions are mathematically consistent and comparable across different machine learning models. Furthermore, SHAP provides both local explanations, which describe the contribution of each feature to an individual prediction, and global explanations, which summarize feature importance across the entire dataset by aggregating SHAP values. Owing to these advantages, SHAP has become a standard interpretability technique in applications involving environmental science, climate modelling, healthcare, and other data-driven disciplines [65]. In this study, SHAP was employed to interpret the predictions of the selected regression model using the Top –20 mRMR-selected meteorological variables. Although PCA was utilized during feature extraction and model development, SHAP analysis was performed on the original selected variables to facilitate direct physical interpretation of the meteorological factors influencing the annual number of HWDs across the eight administrative divisions of Bangladesh. The SHAP analysis reveals in figure 4.12,4.13 and table 4.9 that six of the eight divisions (Barisal, Chittagong, Dhaka, Khulna, Mymensingh, and Rangpur) share a common physical mechanism in which low-level specific humidity (shumP1000) is the most influential predictor. Lower humidity consistently increases SHAP values, indicating that dry boundary-layer conditions promote the occurrence of heat-wave days by reducing evaporative cooling and increasing sensible heat flux. In contrast, Rajshahi and Sylhet exhibit distinct controlling mechanisms. Rajshahi is primarily influenced by 850 hPa geopotential height (zptP850) and 300 hPa temperature (tempP300), suggesting that large-scale anticyclonic circulation and atmospheric subsidence are the dominant drivers of heat-wave formation. Sylhet, however, is controlled mainly by 850 hPa vertical velocity (vvelP850), indicating that convective suppression and vertical atmospheric motion play a more important role than humidity in regulating heat-wave occurrence. Among all divisions, Khulna exhibits the largest SHAP magnitude (approximately -12 to +12), indicating the greatest sensitivity of the prediction model to changes in atmospheric conditions. Dhaka also shows a wide SHAP range, highlighting the dominant influence of near-surface humidity, whereas Rajshahi and Sylhet stand out as dynamically controlled regions with mechanisms that differ from the moisture-driven processes observed across most of Bangladesh. Most of Bangladesh (Dhaka, Khulna, Mymensingh, Rangpur, Barisal, Chittagong) gets hit by heat waves mainly because the air runs dry, while Rajshahi and Sylhet are special cases driven by large-scale air movement (sinking vs. rising air) rather than just moisture.

Chapter 5

Conclusion

This chapter brings together the main outcomes of the study and relates them to its original objectives. It begins with a brief summary of the research, followed by the key findings, study limitations, recommendations for future work, and final concluding remarks.

5.1. Summary of the Study

This study examined the theory and application of PCA as a dimensionality reduction technique for predicting annual HWDs across Bangladesh's eight administrative divisions. The mathematical principles and component-selection criteria of PCA were explained and demonstrated through a small numerical example. The practical analysis used 1940–2025 ERA5 reanalysis data, with mRMR feature selection and PCA applied before training six regression models. Finally, SHAP was used to identify and interpret the atmospheric factors influencing the model predictions. Based on this theoretical and practical framework, the principal findings of the study are summarized below.

5.2. Key Findings

- a) **Spatial pattern.** Heat-wave occurrence follows a clear west-to-east gradient. Rajshahi and Khulna experience the highest average annual HWDs, whereas Sylhet, Chittagong, and Barisal experience the lowest.
- b) **Temporal trends.** No division shows a statistically significant increasing trend during 1940–2025. Khulna and Mymensingh show significant declining trends, while interannual variability dominates the remaining divisions.
- c) **Model performance.** Ridge Regression produced the most stable overall predictions across the eight divisions. PCA features derived from the Top-20 mRMR-selected variables generally achieved the lowest prediction errors.
- d) **SHAP interpretation.** Low-level specific humidity at 1000 hPa is the dominant predictor in six divisions, indicating the importance of dry boundary-layer conditions. Rajshahi is controlled mainly by geopotential height and upper-air temperature, whereas vertical atmospheric motion is most influential in Sylhet.

Although these findings demonstrate the usefulness of the proposed framework, they should be interpreted in light of several study limitations.

5.3. Limitations of the Study

The study was limited by the relatively small annual sample size, which reduced the stability of data-intensive models such as the MLP and GPR. The analysis was restricted to the pre-monsoon season and relied on ERA5 reanalysis rather than direct station observations. In addition, SHAP was applied to the original mRMR-selected variables instead of the PCA-transformed features to preserve physical interpretability. These limitations provide clear directions for extending and strengthening the present research.

5.4. Recommendations for Future Work

Future studies should extend the analysis to all seasons and include additional predictors such as soil moisture, land-use data, and sea surface temperature. Data augmentation or transfer learning could improve the stability of nonlinear models when observations are limited. The framework could also be applied to higher-resolution climate projections to assess future heat-wave risk under different climate scenarios. Together, these recommendations would improve the scope, robustness, and future applicability of the proposed framework. The overall contribution of the study is summarized in the concluding remarks.

5.5. Concluding Remarks

This study demonstrates that combining mRMR feature selection, PCA, and interpretable machine learning provides an effective framework for predicting annual HWDs across Bangladesh. Ridge Regression using PCA features derived from the Top-20 variables produced the most stable overall performance, while SHAP identified the principal atmospheric drivers across the divisions. These findings provide a practical basis for improving regional heat-wave prediction and future climate-risk assessment.

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